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Asymptotic Behaviour of a Class of Stochastic Approximation Procedures*

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Summary. Let $\mathbb{B}$ be a real Banach space. Considering stochastic approximation procedures of Robbins-Monro and Kiefer-Wolfowitz type, one is led to recursion formulas of the form

$$U_{k+1} = U_k - k^{-1} \Gamma_k U_k - k^{-\rho} V_k - k^{-\eta} T_k$$

(ρ and η positive real numbers, U_k, V_k, T_k $\mathbb{B}$-valued random variables, Γ_k random linear operators $\mathbb{B} \to \mathbb{B}$), where – in addition to some further assumptions – $\Gamma_k \to \Gamma$ almost surely and $T_k \to T$, say, almost surely. Letting $(Y_{k,n})_{k \in \{1,\dots,n\}, n \in \mathbb{N}}$ be an array of $\mathbb{B}$-valued random variables such that the partial sums $\sum_{k=1}^{m} V_k$ and $\sum_{k=1}^{m} Y_{k,n}$ do not differ "too much" (uniformly in $m \in \{1,\dots,n\}$), we shall investigate the distance of $(U_m)_{m \in \{1,\dots,n\}}$ and $(U_m^{(n)})_{m \in \{1,\dots,n\}}$, where

$$U_{k+1}^{(n)} = U_k^{(n)} - k^{-1} \Gamma U_k^{(n)} - k^{-\rho} Y_{k,n} - k^{-\eta} T$$

for $k \in \{1,\dots,n-1\}$. The proofs are based on a deterministic argument that enables us to present a unified approach for proving central limit theorems, weak and almost sure invariance principles and variants of the bounded and the functional law of the iterated logarithm for the sequence (U_n).

0. Notations and Assumptions

0.1. General Notations

The following symbols are used throughout the paper: $\mathbb{C}$ = set of all complex numbers, $\mathbb{R}$ = set of all real numbers, $\mathbb{N}$ = set of all positive integers, $\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$, $\mathbb{R}_0^+ = \mathbb{R}^+ \cup \{0\}$, $\mathcal{M}(X, Y)$ = set of all mappings of the set X into

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the set Y, $(\mathbb{B}, \|\cdot\|)$ denotes a real (not necessarily separable) Banach space and $(\mathscr{L}(\mathbb{B},\mathbb{B}), \|\cdot\|)$ the Banach space of all bounded linear operators $\mathbb{B}\to\mathbb{B}$ equipped with the usual norm, $I\in\mathscr{L}(\mathbb{B},\mathbb{B})$ is the identity operator, and $\chi(A)$ is the indicator function of a set A. For $x\in\mathbb{R}^+$, we use $\mathrm{L}_2 x$ to mean $\log\log(\max(e^e, x))$.

The phrase "without loss of generality, there exist random variables ... such that" is to be understood in Strassen's [50] sense, namely: "without changing their distribution, one can redefine the random variables ... on a new probability space such that".

0.2. Description of the Basic Model

Let $(\Omega, \mathscr{A}, P)$ be a probability space. Let β, $\delta\in\mathbb{R}_0^+$, $T\in\mathscr{M}(\Omega,\mathbb{B})$, $\Gamma\in\mathscr{L}(\mathbb{B},\mathbb{B})$ and $\gamma\in\mathbb{R}^+$. For $n\in\mathbb{N}$, let T_n, U_n, $V_n\in\mathscr{M}(\Omega,\mathbb{B})$, $\Gamma_n\in\mathscr{M}(\Omega,\mathscr{L}(\mathbb{B},\mathbb{B}))$ and suppose

$$U_{n+1}=(I-n^{-1}\Gamma_n)\,U_n+n^{-(1+\beta)/2}\,V_n+n^{-1-\beta/2}\,T_n. \tag{0.2.C0}$$

Also assume

$$2\gamma>\beta, \tag{0.2.C1}$$

$$\|e^{-u\Gamma}\|<e^{-\gamma u} \quad \text{for all } u\in\mathbb{R}^+, \tag{0.2.C2}$$

and suppose there exists a P-null set $M_0\in\mathscr{A}$ such that for all $\omega\in\Omega\setminus M_0$ the following condition is satisfied:

$$\lim_{n\to\infty} n^{\delta}\,\|\Gamma_n(\omega)-\Gamma\|=0. \tag{0.2.C3}$$

0.3. Remarks Concerning 0.2

Remark 0.1. In the sequel, we shall need at least two further conditions to specify the complete models. Loosely stated, the first will be that the partial sums of the sequence (V_n) satisfy an invariance principle like result, and the second that – for large n – most of the T_n's are in an appropriate sense close to T.

Remark 0.2. (0.2.C2) implies that the spectrum of the operator $\tilde{\Gamma}$ obtained from Γ by linear extension to the complexification of $\mathbb{B}$ is contained in $\{z\in\mathbb{C}\colon \operatorname{Re} z>\gamma\}$. Conversely, if the spectrum of $\tilde{\Gamma}$ is contained in $\{z\in\mathbb{C}\colon \operatorname{Re} z>\gamma\}$, then it is possible to find an equivalent norm on $\mathbb{B}$ such that (0.2.C2) holds with respect to the induced norm on $\mathscr{L}(\mathbb{B},\mathbb{B})$. Cf. Daleckiĭ and Kreĭn [8], p. 29f., and Walk [52], [55].

1. Introduction

The model described in 0.2 constitutes a natural framework for treating questions concerning the validity of the central limit theorem (CLT), weak and almost sure invariance principles and the law of the iterated logarithm (LIL) for a wide class of stochastic approximation procedures including the Robbins-Monro process, the Kiefer-Wolfowitz process and Venter's design (cf. Fabian [12], Walk [52], Mark [32] and Nixdorf [39], [40]). This idea goes back to

Fabian[1] [12] who investigated the asymptotic normality of the sequence (U_n) in the case $\mathbb{B}=\mathbb{R}^d$ with (V_n) being a martingale difference sequence.

The present paper was motivated by the following two observations:

(a) If $\Gamma_n \to \Gamma$ a.s. and $T_n \to T$ in a suitable sense, it can be shown that under some additional hypotheses the sequence (U_n^*) obtained from (U_n) by replacing the sequences (Γ_n) and (T_n) in (0.2.C0) by the constant sequences taking only the values Γ and T, resp., is close to the original sequence in the sense that $n^{\beta/2}(U_n - U_n^*)$ converges to 0 in probability (cf. Fabian [12], [15]). (This can be utilized to deduce a CLT for $(n^{\beta/2} U_n)$ from a CLT for the simpler sequence $(n^{\beta/2} U_n^*)$.)

(b) Investigating stochastic approximation procedures, several authors (Walk [52], Lai and Robbins [28], Ruppert [46], Mark [32]) successfully exploited the fact that it is possible to derive certain properties (weak and almost sure invariance principles, LIL) for the sequence (U_n) from related properties of the partial sum sequence $\left(\sum_{k=1}^{n} V_k\right)_{n\in\mathbb{N}}$ by using partial summation.

The ideas indicated in (a) and (b) suggest that there should be a general theorem as to the stability of the stochastic difference equation (0.2.C0) stating that the solutions are in some sense insensitive with regard to "small" perturbations of the Γ_n's, T_n's and the partial sums $\sum_{k=1}^{n} V_k$. A result of this type is provided by Proposition 2.1 below. It is of a purely deterministic nature. Taking some well-known theorems for martingales and partial sums of weakly dependent random variables for granted, results like CLT, LIL, weak and almost sure invariance principles for the sequence (U_n) are almost immediate consequences of this proposition. The results obtained in this way include or generalize the CLT's of Fabian [12] (case $\alpha=1$) and Borodin [6] (discrete case), LIL-type results due to Gaposhkin and Krasulina [18], Heyde [21] and Mark [32], weak invariance principles published by McLeish [34] and Walk [52], and strong approximation theorems proved by Kersting [23], Ruppert [46] and Mark [32]. Because of this variety of applications, the present article may also be useful as a survey of a certain range of work done in stochastic approximation.

2. Main Results

Before we state our main results, we need to introduce some more notation. Let γ and Γ be as in Sect. 0.2, and let $\rho\in\mathbb{R}_0^+$. For $(k,n)\in\mathbb{N}^2$ with $k\leqq n$, let

$$v_{k,n},\ z_{k,n}\in\mathbb{B} \quad\text{and}\quad \gamma_n\in\mathscr{L}(\mathbb{B},\mathbb{B}). \tag{2.1}$$

For $N\in\mathbb{N}$, $(k,n)\in\mathbb{N}^2$ with $k<n\leqq N$, $l\in\mathbb{N}$, $i\in\{0,1\}$ and $j\in\{0,1,2,3\}$, put

$$\Gamma_l^{(0)}=\gamma_l, \qquad \Gamma_l^{(1)}=\Gamma, \tag{2.2}$$

1 Actually, Fabian considered a more general recursion formula of the form

$$U_{n+1}=(I-n^{-\alpha}\Gamma_n)\,U_n+n^{-(\alpha+\beta)/2}\,V_n+n^{-\alpha-\beta/2}\,T_n$$

with $\alpha\in(0,1]$. Basically, the case $\alpha\in(0,1)$ can also be treated by the methods of this, paper but the details are somewhat different.

and define

$$A_{k,n}^{(i)}=(I-n^{-1}\Gamma_n^{(i)})\cdot\ldots\cdot(I-(k+1)^{-1}\Gamma_{k+1}^{(i)}), \tag{2.3}$$

$$A_{k,n}^{(2)}=\exp\left(-\left(\sum_{p=k+1}^{n}p^{-1}\right)\Gamma\right),\qquad A_{k,n}^{(3)}=\left(\frac{k}{n}\right)^{\Gamma},$$

$$A_{l,l}^{(j)}=I,$$

$$C_k^{(i)}=I-(k+1)^{-1}\Gamma_{k+1}^{(i)},\qquad C_k^{(2)}=\exp(-(k+1)^{-1}\Gamma), \tag{2.4}$$

$$C_k^{(3)}=\left(\frac{k}{k+1}\right)^{\Gamma},\qquad B_k^{(j)}=k^{-\rho}C_k^{(j)}-(k+1)^{-\rho}I,$$

$$u_{n,N}^{(j)}=\sum_{p=1}^{n}A_{p,n}^{(j)}p^{-\rho}v_{p,N},\qquad y_{n,N}=\sum_{p=1}^{n}A_{p,n}^{(3)}p^{-\rho}z_{p,N}. \tag{2.5}$$

Let $N\in\mathbb{N}$ be arbitrary, write

$$I_N=\{1,\ldots,N\},\qquad \mathbb{G}_N=\{(k,n)\in\mathbb{N}^2: k\leqq n\leqq N\},$$
$$\mathscr{P}_N=\{S: S\subset\{1,\ldots,N\}, S\neq\emptyset\},$$

and let the mappings

$$C_N\colon \mathscr{P}_N\times\mathscr{M}(\mathbb{G}_N,\mathscr{L}(\mathbb{B},\mathbb{B}))\times\mathscr{M}(I_N,\mathbb{R}^+)\to\mathbb{R}_0^+$$

and

$$\Delta_N\colon \mathscr{P}_N\times\mathscr{M}(I_N,\mathbb{B})\times\mathscr{M}(I_N,\mathbb{R}^+)\to\mathbb{R}_0^+$$

be defined by

$$C_N(J,A,\lambda)=\max_{n\in J}\left(\lambda(n)\,\|A(n,n)\|+\sum_{j=1}^{n-1}\lambda(j)\,\|A(j,n)-A(j+1,n)\|\right)$$

and

$$\Delta_N(J,x,\lambda)=\max_{n\in J}\max_{1\leqq l\leqq n}\lambda(l)^{-1}\left\|\sum_{j=1}^{l}x(j)\right\|.$$

Proposition 2.1. *Let $J\in\mathscr{P}_N$, $\delta\in[0,1)$, $\theta,\tau\in\mathbb{R}_0^+$, put $\kappa=\theta+\gamma-\rho-\delta$, and write – for $(k,n)\in\mathbb{G}_N$ –*

$$v^{(N)}(k)=v_{k,N},\qquad z^{(N)}(k)=z_{k,N},\qquad \lambda(k)=\left(\frac{k}{N}\right)^{\theta},$$

$$A'(k,n)=n^{\tau}k^{-\rho}(A_{k,n}^{(0)}-A_{k,n}^{(3)}),\qquad A''(k,n)=n^{\tau}k^{-\rho}A_{k,n}^{(3)}.$$

If (Γ,γ) satisfies (0.2.C2) *and*

$$\lim_{n\to\infty}n^{\delta}\,\|\gamma_n-\Gamma\|=0, \tag{2.1.C3}$$

then the following statements hold true:

(a)
$$\max_{n\in J}n^{\tau}\,\|u_{n,N}^{(0)}-y_{n,N}\|\leqq C_N(J,A',\lambda)\,\Delta_N(J,v^{(N)},\lambda)+C_N(J,A'',\lambda)\,\Delta_N(J,v^{(N)}-z^{(N)},\lambda).$$

(b) *The C_N's can be estimated from above by*

$$C_N(J,A',\lambda)\leqq\max_{n\in J}\left(\frac{n}{N}\right)^{\theta}a_n\,n^{\tau-\rho-\delta}\,d_n(\kappa)\int_{1/n}^{1}t^{\kappa-1}\,dt \tag{2.6}$$

and

$$C_N(J, A'', \lambda) \leqq c_{10} \max_{n \in J} \left(\frac{n}{N}\right)^{\theta} n^{\tau-\rho} \int_{1/n}^{1} t^{\theta+\gamma-\rho-1}\, dt, \tag{2.7}$$

where

$$c_{10} \in \mathbb{R}^+ \quad \textit{is a constant}, \tag{2.8}$$

$$d_n(\kappa) = \begin{cases} O(1) & \textit{if } \kappa \leqq 0 \\ o(1) & \textit{if } \kappa > 0 \end{cases} \quad (\textit{as } n \to \infty), \tag{2.9}$$

and – with $g(1,n) = \sum_{l=2}^{n} l^{-1} \|\gamma_l - \Gamma\|$ *and* $h(1,n) = \exp(g(1,n))$ –

$$a_n = \begin{cases} 1 & \textit{if } \max(\kappa, \delta) > 0 \\ 1 + g(1,n)\, h(1,n) & \textit{if } \max(\kappa, \delta) \leqq 0. \end{cases} \tag{2.10}$$

Proof. See Sect. 4.

In the sequel, P^* denotes the outer measure corresponding to P. A proposition concerning the points of Ω is said *to hold true almost surely* (a.s.) if it holds for all $\omega \in \Omega \setminus A$, where $A \subset \Omega$ is a set with $P^*(A) = 0$. A sequence $(X_n)_{n \in \mathbb{N}}$ in $\mathscr{M}(\Omega, \mathbb{B})$ is said *to converge in probability* to a mapping X_∞ in $\mathscr{M}(\Omega, \mathbb{B})$ (or $X_n \xrightarrow{P} X_\infty$) if $\lim_{n \to \infty} P^*\{\|X_n - X_\infty\| \geqq \varepsilon\} = 0$ for every $\varepsilon \in \mathbb{R}^+$ (cf. Dudley and Philipp [11], p. 511). (X_n) is said *to converge to* X_∞ *almost in probability* (or $X_n \xrightarrow{P^*} X_\infty$) if there is a family $\{A_{n,\varepsilon}\colon n \in \mathbb{N},\ \varepsilon \in (0,1)\}$ of subsets of Ω such that $P^*(A_{n,\varepsilon}) \leqq \varepsilon$ (for each $\varepsilon \in (0,1)$ and all $n \in \mathbb{N}$) and $\lim_{n \to \infty} \|X_n - X_\infty\|\, \chi(\Omega \setminus A_{n,\varepsilon}) = 0$ pointwise. The last definition will turn out to be rather useful. A partial explanation of this fact is provided by the following (easily verified) properties:

(i) $X_n \xrightarrow{P} X_\infty$ implies $X_n \xrightarrow{P^*} X_\infty$.

(ii) If the mappings $\omega \mapsto \|(X_n - X_\infty)(\omega)\|$ are measurable, then

$$X_n \xrightarrow{P^*} X_\infty \quad \text{is equivalent to} \quad X_n \xrightarrow{P} X_\infty.$$

(iii) Without measurability assumptions, it is not true in general that $X_n \to X_\infty$ a.s. (or even pointwise) implies

$$X_n \xrightarrow{P} X_\infty; \text{ however, it implies } X_n \xrightarrow{P^*} X_\infty.$$

Theorem 2.2. *Let the hypotheses introduced in Sect.* 0.2 *be satisfied with* $\delta = 0$. *Let* $\theta \in \left(\max\left(0, \frac{1+\beta}{2} - \gamma\right), \frac{1}{2}\right)$, *and assume*

$$\lim_{r \to \infty} \limsup_{N \to \infty} P^* \left\{ N^{-1/2} \max_{n \leqq N} \left(\frac{N}{n}\right)^{\theta} \left\| \sum_{k=1}^{n} V_k \right\| \geqq r \right\} = 0, \tag{2.2.C4}$$

$$N^{-1/2} \max_{n \leqq N} \left(\frac{N}{n}\right)^{\theta} \left\| \sum_{k=1}^{n} k^{-1/2} (T_k - T) \right\| \xrightarrow{P^*} 0 \quad \textit{as } N \to \infty. \tag{2.2.C5}$$

Let $(Z_{n,N})_{(n,N)\in\mathbb{N}^2, n\leqq N}$ *be an array in* $\mathscr{M}(\Omega,\mathbb{B})$ *such that*

$$\text{(2.2.C6)}\qquad N^{-1/2}\max_{n\leqq N}\left(\frac{N}{n}\right)^{\theta}\left\|\sum_{k=1}^{n}(V_k-Z_{k,N})\right\|\xrightarrow{P^*}0\quad \textit{as } N\to\infty.$$

Then – writing

$$\text{(2.11)}\qquad Y_{n+1,N}=\sum_{k=1}^{n}\left(\frac{k}{n}\right)^{\Gamma}k^{-\frac{1+\beta}{2}}Z_{k,N}-n^{-\frac{\beta}{2}}\left(\Gamma-\frac{\beta}{2}I\right)^{-1}T$$

for $(n,N)\in\mathbb{N}^2$ *with* $n\leqq N$ *– one has*

$$\text{(2.12)}\qquad N^{-1/2}\max_{n\leqq N}n^{(1+\beta)/2}\|U_{n+1}-Y_{n+1,N}\|\xrightarrow{P^*}0\quad \textit{as } N\to\infty.$$

Proof. Let $\omega\in\Omega$ be arbitrarily fixed, and choose $\gamma_k=\Gamma_k(\omega)$ $(k\in\mathbb{N})$. By induction, it follows from (0.2.C0) that

$$U_{n+1}(\omega)=\sum_{j=1}^{3}U_{n+1}^{(j)}(\omega),$$

where

$$\text{(2.13)}\qquad \begin{aligned}U_{n+1}^{(1)}(\omega)&=(I-n^{-1}\gamma_n)\cdot\ldots\cdot(I-\gamma_1)\,U_1(\omega),\\ U_{n+1}^{(2)}(\omega)&=\sum_{k=1}^{n}A_{k,n}^{(0)}k^{-\frac{1+\beta}{2}}V_k(\omega),\\ U_{n+1}^{(3)}(\omega)&=\sum_{k=1}^{n}A_{k,n}^{(0)}k^{-1-\frac{\beta}{2}}T_k(\omega).\end{aligned}$$

We also have

$$Y_{n+1,N}(\omega)=\sum_{j=1}^{3}Y_{n+1,N}^{(j)}(\omega),$$

where

$$\text{(2.14)}\qquad \begin{aligned}Y_{n+1,N}^{(1)}(\omega)&=\sum_{k=1}^{n}A_{k,n}^{(3)}k^{-\frac{1+\beta}{2}}Z_{k,N}(\omega),\\ Y_{n+1,N}^{(2)}(\omega)&=\sum_{k=1}^{n}A_{k,n}^{(3)}k^{-1-\frac{\beta}{2}}T(\omega),\\ Y_{n+1,N}^{(3)}(\omega)&=n^{-\beta/2}\left(\Gamma-\frac{\beta}{2}I\right)^{-1}T(\omega)-Y_{n+1,N}^{(2)}(\omega).\end{aligned}$$

(Note that, in view of (0.2.C1) and Remark 0.2, $\Gamma-\frac{\beta}{2}I$ is invertible.) To prove (2.12) it is enough to show that the mappings $U_{n+1}^{(j)}$, $Y_{n+1,N}^{(j)}$ $(j\in\{1,2,3\})$ defined by (2.13) and (2.14) satisfy

$$\text{(2.15)}\qquad N^{-1/2}\max_{n\leqq N}n^{(1+\beta)/2}\|U_{n+1}^{(1)}\|\xrightarrow{P^*}0\quad \text{as } N\to\infty,$$

$$\text{(2.16)}\qquad N^{-1/2}\max_{n\leqq N}n^{(1+\beta)/2}\|U_{n+1}^{(j+1)}-Y_{n+1,N}^{(j)}\|\xrightarrow{P^*}0\quad \text{as } N\to\infty\quad \text{for } j\in\{1,2\},$$

$$\text{(2.17)}\qquad N^{-1/2}\max_{n\leqq N}n^{(1+\beta)/2}\|Y_{n+1,N}^{(3)}\|\xrightarrow{P^*}0\quad \text{as } N\to\infty.$$

(2.15) is an immediate consequence of (0.2.C3) and Lemma 3.3 below. The proof of (2.16) will be carried out only in the case $j=1$. (The case $j=2$ can be treated analogously.) Applying Proposition 2.1 with

$$J=\{1,\ldots,N\},\ v_{k,N}=V_k(\omega),\ z_{k,N}=Z_{k,N}(\omega),\ \lambda(m)=\left(\frac{m}{N}\right)^{\theta},$$

$$u_{n,N}^{(0)}=U_{n+1}^{(2)}(\omega),\ y_{n,N}=Y_{n+1,N}^{(1)}(\omega),\ \tau=\rho=(1+\beta)/2,\ \delta=0$$

and using the same notation as there, we find that – for $\omega\in\Omega\setminus M_0$ (with M_0 being defined in Sect. 0.2) –

$$C_N(J,A',\lambda)=o(1)\ (\text{as } N\to\infty)\ \text{and}\ C_N(J,A'',\lambda)=O(1) \tag{2.18}$$

(since $\kappa>0$ and $a_n=1$ for all n). Combining (2.18),

$$\Delta_N(J,v^{(N)},\lambda)=\max_{n\leqq N}\left(\frac{N}{n}\right)^{\theta}\left\|\sum_{k=1}^{n}V_k(\omega)\right\|, \tag{2.19}$$

$$\Delta_N(J,v^{(N)}-z^{(N)},\lambda)=\max_{n\leqq N}\left(\frac{N}{n}\right)^{\theta}\left\|\sum_{k=1}^{n}(V_k(\omega)-Z_{k,N}(\omega))\right\|, \tag{2.20}$$

(2.2.C4) and (2.2.C6), the case $j=1$ of (2.16) is implied by Proposition 2.1 (a). A sufficient condition for (2.17) to hold is

$$\lim_{n\to\infty}\left\|\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^{\Gamma-\left(1+\frac{\beta}{2}\right)I}-\left(\Gamma-\frac{\beta}{2}I\right)^{-1}\right\|=0, \tag{2.21}$$

which in turn follows from the equality

$$\int_0^1 t^{\Gamma-\left(1+\frac{\beta}{2}\right)I}\,dt=\left(\Gamma-\frac{\beta}{2}I\right)^{-1}$$

and a routine argument based on the (Banach space) mean value theorem or Lebesgue's dominated convergence theorem for Bochner integrals.

Remark 2.3. Suppose that

(2.3.C5) for every $\varepsilon\in\mathbb{R}^+$ there exist a set $\Omega_\varepsilon\in\mathscr{A}$ with $P(\Omega_\varepsilon)\geqq 1-\varepsilon$ and a sequence $(\xi_n^{(\varepsilon)})_{n\in\mathbb{N}}$ of nonnegative, real-valued random variables such that

$$\frac{1}{n}\sum_{k=1}^{n}\|(T_k-T)(\omega)\|\leqq\xi_n^{(\varepsilon)}(\omega)\qquad\text{for all } \omega\in\Omega_\varepsilon,\quad n\in\mathbb{N}$$

and

$$\lim_{n\to\infty}E(\xi_n^{(\varepsilon)})=0.$$

Then it follows from the obvious inequality

$$\max_{n\leqq N}\left(\frac{N}{n}\right)^{\theta}\sum_{k=1}^{n}k^{-1/2}\|T_k-T\|\leqq\sum_{k=1}^{N}k^{-1/2}\|T_k-T\|$$
$$+\sum_{n=1}^{N-1}\left(\left(\frac{N}{n}\right)^{\theta}-\left(\frac{N}{n+1}\right)^{\theta}\right)\sum_{k=1}^{n}k^{-1/2}\|T_k-T\|$$

together with the relation

$$\sum_{k=1}^{n} k^{-1/2} \|T_k - T\| = n^{-1/2} \sum_{k=1}^{n} \|T_k - T\| + \sum_{k=1}^{n-1} O(k^{-3/2}) \sum_{j=1}^{k} \|T_k - T\|$$

(partial summation) and the Toeplitz lemma that condition (2.2.C5) is fulfilled; especially (2.2.C5) is weaker than "convergence almost in first mean" of (T_n) to T (cf. Walk [52], p. 139).

Remark 2.4. Theorem 2.2 includes a sharpened version of the case $\alpha=1$ of Fabian's [15] Lemma 2. To prove this, we use the same notation as in [15]. Replacing the Φ_n's and V_n's in Fabian's lemma by $\tilde{\Phi}_n$'s and $\tilde{V}_n$'s defined by

$$\tilde{\Phi}_n = \Phi_n \chi(\{\|\Phi_n\| \leqq \|\Phi\| + 1\})$$

and

$$\tilde{V}_n = V_n \chi(\{E(\|V_n\|^2 | \mathscr{F}_n) \leqq C_0 + 1\}),$$

where $C_0 = \operatorname{ess\,sup} \limsup_{n\to\infty} E(\|V_n\|^2 | \mathscr{F}_n)$, it follows from Lemma 3.1 below that – for each $\theta \in (0, 1/2)$ –

(2.22) (2.2.C4) and (2.2.C6) hold with $\tilde{\Phi}_k \tilde{V}_k$ in place of V_k and $\Phi \tilde{V}_k$ in place of $Z_{k,N}$.

By the definition of C_0 and since $\Phi_n \to \Phi$ a.s. as $n \to \infty$, (2.22) still remains true if the tildes are omitted. This together with Remark 2.3 completes the proof.

Remark 2.5. Let $\mathbb{H}$ be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$ ($\|x\| = \langle x, x \rangle^{1/2}$). Let $\mathbb{H}$ be endowed with its Borel σ-field. Let $(V_n)_{n\in\mathbb{N}}$ be a sequence of $\mathbb{H}$-valued random variables defined on the probability space $(\Omega, \mathscr{A}, P)$, and let $(\mathscr{F}_n)_{n\in\mathbb{N}\cup\{0\}}$ be a sequence of sub-σ-fields of $\mathscr{A}$ such that – for each $n \in \mathbb{N}$ – V_n is $\mathscr{F}_n$-measurable and $\mathscr{F}_{n-1} \subset \mathscr{F}_n$. Let S be as in Theorem 5.1 infra, and suppose that

(2.5.C6) the hypotheses of Theorem 5.1 are satisfied with

$$X_{n,i} = n^{-1/2} V_i, \quad \mathscr{F}_{n,i} = \mathscr{F}_i \text{ and } k_n(t) = [nt]$$
$$((n,i) \in \mathbb{N}^2, i \leqq n, t \in [0,1]).$$

Let $\theta \in [0, 1/2)$ be arbitrary. Then (2.2.C4) holds and, without loss of generality, there exist $\mathbb{H}$-valued normally distributed random variables $Z_{k,N}$ ($k \in \{1, \ldots, N\}$, $N \in \mathbb{N}$) with $EZ_{k,N} = 0$ and $\operatorname{Cov}(Z_{k,N}) = S$ such that – for each N – $Z_{1,N}, \ldots, Z_{N,N}$ are independent and (2.2.C6) is satisfied.

Proof. In view of the argument used at the beginning of the proof of Theorem 5.1 below, it is obvious that we may confine ourselves to proving the assertion of Remark 2.5 under the additional assumption

$$\sup_{n\in\mathbb{N}} \frac{1}{n} \sum_{k=1}^{n} E(\|V_k\|^2 | \mathscr{F}_{k-1}) \leqq c \quad \text{a.s. for some } c \in \mathbb{R}^+. \tag{2.23}$$

In this case it follows from Lemma 3.1 below that – for all $M \in \{1, \ldots, N\}$, $N \in \mathbb{N}$, $r \in \mathbb{R}^+$, $\theta \in [0, 1/2)$ –

$$P\left\{\max_{n \leqq M} \left(\frac{N}{n}\right)^{\theta} \left\| \sum_{k=1}^{n} V_k \right\| \geqq r N^{1/2}\right\} \leqq c_{11} r^{-1} \left(\frac{M}{N}\right)^{1/2-\theta}, \tag{2.24}$$

where c_{11} is a positive constant not depending on r, M and N. Choosing $M = N$, this implies (2.2.C4). Combining Theorem 5.1, the Skorohod-Dudley theorem (see [10], Theorem 3) and Lemma 2.13 in [11], we find that the conclusion of Remark 2.5 holds for $\theta = 0$. But together with (2.24) (take $M = [\eta N]$ with $\eta \in (0, 1)$) and a corresponding estimate for

$$P\left\{\max_{n \leqq M} \left(\frac{N}{n}\right)^{\theta} \left\| \sum_{k=1}^{n} Z_{k,N} \right\| \geqq r N^{1/2}\right\},$$

this also entails the desired result for arbitrary $\theta \in [0, 1/2)$.

Corollary 2.6. *Under the basic hypotheses of Theorem 2.2, let* $(\mathbb{B}, \|\cdot\|) = (\mathbb{H}, \|\cdot\|)$ *be a real separable Hilbert space, let* (V_n) *be chosen as in Remark 2.5, and let (2.2.C4) and (2.2.C6) be replaced by (2.5.C6). Then (2.12) is satisfied for the array* $(Z_{k,N})_{k \in \{1, \ldots, N\}, N \in \mathbb{N}}$ *described in Remark 2.5.*

Proof. Obvious.

Corollary 2.7. *Let the hypotheses of Corollary 2.6 be fulfilled. Then, without loss of generality, there exists a sequence of* $\mathbb{H}$*-valued Brownian motions* W_N *with* $W_N(0) = 0$ *and* $\operatorname{Cov}(W_N(1)) = S$ *such that – with* $\Lambda = \Gamma - \frac{1+\beta}{2} I$ *and*

$$\hat{Y}_N(t) = W_N(t) - \Lambda \int_0^1 s^{\Lambda - I} W_N(st)\, ds - t^{1/2} \left(\Gamma - \frac{\beta}{2} I\right)^{-1} T - \tag{2.25}$$

$$N^{-1/2} \max_{0 \leqq t \leqq N} \|[t]^{(1+\beta)/2} U_{[t+1]} - \hat{Y}_N(t)\| \xrightarrow{P^*} 0 \quad \text{as } N \to \infty. \tag{2.26}$$

Proof. Put $\kappa = \theta + \gamma - \frac{1+\beta}{2}$. In view of our choice of θ, κ is positive. We first recall that the integral on the right-hand side of (2.25) is well-defined on a set of P-measure 1 (not depending on t); indeed – for $t \in [0, N]$ –

$$\begin{aligned} \int_0^1 \|s^{\Lambda - I}\| \, \|W_N(st)\| \, ds &\leqq t^{\theta} \int_0^1 s^{\kappa - 1} (st)^{-\theta} \|W_N(st)\| \, ds \qquad \text{(by (0.2.C2))} \\ &\leqq N^{\theta} \kappa^{-1} \sup_{0 < u \leqq N} u^{-\theta} \|W_N(u)\| < \infty \\ &\qquad \text{a.s. (by Lemma 3.2).} \end{aligned}$$

Let us now turn to the proof of (2.26). The method of obtaining the array $(Z_{k,N})$ in the proof of Remark 2.5 actually yields a sequence of Brownian motions W_N such that $W_N(n) = \sum_{k=1}^{n} Z_{k,N}$ for $n \in \{1, \ldots, N\}$. Hence – taking these

W_N's in (2.25) and using Corollary 2.6 – it remains to show that

$$N^{-1/2} \max_{0 \leqq t \leqq N} \|[t]^{(1+\beta)/2} Y_{[t+1],N} - \hat{Y}_N(t)\| \xrightarrow{P} 0 \quad \text{as } N \to \infty, \tag{2.27}$$

where $(Y_{n+1,N})$ is defined as in (2.11). Writing

$$K(N) = \max_{k \in \{1,\ldots,N-1\}} \max_{k-1 \leqq t \leqq k+1} \left(\frac{N}{k}\right)^{\theta} \|W_N(t) - W_N(k)\|$$

one has

$$K(N) = O(N^{\theta}) \quad \text{in probability} \tag{2.28}$$

(by Fernique's [16] theorem and the Borel-Cantelli lemma), so that (2.27) follows from the estimates

$$\max_{t \in \mathbb{R}_0^+} |t^{1/2} - [t+1]^{1/2}| = 1, \tag{2.29}$$

$$\begin{aligned} &\max_{n \in \{1,\ldots,N\}} \left\| \sum_{k=1}^{n} \left(\frac{k}{n}\right)^{\Lambda} Z_{k,N} - W_N(n) + \Lambda \int_{1/n}^{1} s^{\Lambda - I} W_N(sn)\, ds \right\| \\ &= \max_{n \in \{1,\ldots,N\}} \left\| \sum_{k=1}^{n} \left(\left(\frac{k}{n}\right)^{\Lambda} - \left(\frac{k+1}{n}\right)^{\Lambda} \right) W_N(k) + \Lambda \int_{1/n}^{1} s^{\Lambda - I} W_N(sn)\, ds \right\| \\ &\qquad\qquad \text{(by partial summation)} \\ &= \max_{n \in \{1,\ldots,N\}} \left\| \Lambda \sum_{k=1}^{n-1} \int_{k/n}^{(k+1)/n} s^{\Lambda - I} (W_N(k) - W_N(sn))\, ds \right\| \\ &\leqq \|\Lambda\| \int_0^1 s^{\kappa - 1}\, ds\, K(N) \qquad \text{(by (0.2.C2))} \end{aligned} \tag{2.30}$$

and

$$\begin{aligned} &\max_{n \in \{1,\ldots,N\}} \max_{n-1 \leqq t < n} \left\| W_N(t) - \Lambda \int_0^1 s^{\Lambda - I} W_N(st)\, ds \right. \\ &\qquad \left. - W_N(n) + \Lambda \int_{1/n}^{1} s^{\Lambda - I} W_N(sn)\, ds \right\| \\ &\leqq \max_{n \in \{1,\ldots,N\}} \max_{n-1 \leqq t < n} \|W_N(t) - W_N(n)\| \\ &\qquad + \|\Lambda\| \int_0^1 s^{\kappa - 1}\, ds \left(2K(N) + N^{\theta} \sup_{0 < u \leqq 1} u^{-\theta} \|W_N(u)\|\right) \\ &= O(N^{\theta}) \text{ in probability} \\ &\qquad \text{(using Fernique's theorem, the Borel-Cantelli lemma,} \\ &\qquad \text{(2.28) and Lemma 3.2).} \end{aligned} \tag{2.31}$$

Remark 2.8. Suppose, in addition to the hypotheses of Corollary 2.7, that – for all $n \in \mathbb{N}$ – the mappings T_n, U_n and T are Borel measurable. Then – using the path-continuity of the process Y_1^* defined below – the assertion of Corollary 2.7 can be reformulated as follows:

The sequence of random elements $\hat{U}_N$ in the Banach space $C_{\mathbb{H}}[0,1]$ of $\mathbb{H}$-valued continuous functions on $[0,1]$ (endowed with the sup-norm) given by

$$\hat{U}_N(t) = N^{-1/2}\{R_{[Nt]} + (Nt - [Nt])(R_{[Nt]+1} - R_{[Nt]})\} \qquad (t \in [0,1])$$

with $R_n = n^{(1+\beta)/2} U_{n+1} - n^{1/2}\left(\Gamma - \frac{\beta}{2} I\right)^{-1} T$ converges in distribution to the process Y_1^* defined by

$$Y_1^*(t) = W_1(t) - \Lambda \int_0^1 s^{\Lambda - I} W_1(st)\, ds \qquad (t \in [0,1]).$$

The functional CLT stated in Remark 2.8 comprises the following results as special cases:

(a) The case $\alpha = 1$ *of Fabian's CLT* ([12], *Theorem* 2.2). Fabian's recursion formula ((2.26) in [12]) differs from (0.2.C0) in that there appear terms of the form $\Phi_n V_n$ instead of V_n. The Φ_n's are random linear operators satisfying $\lim_{n\to\infty} \Phi_n = \Phi$ a.s. (for some nonrandom linear operator Φ) and a measurability condition. Because of a misprint (cf. Fabian [15], p. 6), this measurability condition should be read as "Φ_n is $\mathscr{F}_n$-measurable". But then the sequence $(\Phi_n V_n, \mathscr{F}_{n+1})$ in Fabian's theorem is a martingale difference sequence subject to the conditions imposed on the sequence $(V_n, \mathscr{F}_n)$ in Remark 2.5 (and Remark 2.8).

(b) Walk's functional CLT ([52], *Theorem* 1). Under technical assumptions slightly more restrictive than those of the present result, Walk considered the special case $\beta = 1$, $\Gamma_n = \Gamma$, $T = 0$ of (0.2.C0) and derived a functional CLT ([52], Theorem 3) for the Robbins-Monro process from his theorem. (The formulation of Walk's Theorem 1 contains a little inaccuracy: The conditioning with respect to $X_1, \ldots, X_n$ should always be replaced by a conditioning with respect to a σ-field $\mathscr{G}_n$ such that – for all $n \in \mathbb{N}$ – V_n is $\mathscr{G}_{n+1}$-measurable and $\mathscr{G}_n \subset \mathscr{G}_{n+1}$.) We also mention that – exploiting the fact that a function $f\colon \mathbb{H} \to \mathbb{H}$ is Fréchet differentiable at 0 iff it can be represented in the form $f(x) = \Delta(x)\, x$ $(x \in \mathbb{H})$, where $\Delta \in \mathscr{M}(\mathbb{H}, \mathscr{L}(\mathbb{H}, \mathbb{H}))$ is continuous at 0 – Walk's Theorem 3 follows from Remark 2.8 without any further computations. This shows that it is indeed advantageous to use a recursion formula with a sequence of random linear operators Γ_n such that $\Gamma_n \to \Gamma$ a.s. instead of a fixed operator Γ.

(c) Variants of Walk's theorem due to Pantel ([42], *Satz* 8.2) *and the author* [2] (*see* [39], *Satz* 2.1.1, *or* [40]). Pantel (for $\mathbb{H} = \mathbb{R}^d$) and the present author (for general $\mathbb{H}$) have independently obtained versions of Walk's theorem which include the case $\alpha = 1$ of Fabian's theorem ([12], Theorem 2.2). Pantel applied his theorem for proving limit theorems for adaptive stochastic approximation procedures in $\mathbb{R}^d$ (i.e., multivariate analogues of Venter's [51] design). The result in [2] has been utilized by Nixdorf [39], [40] in order to derive functional CLT's for Robbins-Monro and Kiefer-Wolfowitz type algorithms in a Hilbert space.

For further possible applications of Corollary 2.7 and Remark 2.8, the reader is referred to Abdelhamid [1], Fabian [13], [14], [15], Lai and Robbins [29] and Nevel'son and Has'minskiĭ [38], § VI, 6.

Let $\mathbb{B}$ be an arbitrary real Banach space again. For $\theta \in [0, 1/2)$, let $C^\theta_{\mathbb{B}}[0,1]$ denote the linear space of all continuous functions f: $[0,1] \to \mathbb{B}$ such that $\sup_{t\in(0,1]} t^{-\theta} \|f(t)\| < \infty$. Equipped with the norm $\|\cdot\|_\theta$ defined by $\|f\|_\theta = \sup_{t\in(0,1]} t^{-\theta}\|f(t)\|$, $C^\theta_{\mathbb{B}}[0,1]$ is a Banach space. Moreover, if $\mathbb{B}$ is separable, then $(C^\theta_{\mathbb{B}}[0,1], \|\cdot\|_\theta)$ is separable, too. The norms $\|\cdot\|_\theta$ just introduced are a particular case of the so-called q-norms (cf., e.g., Pyke [45] and O'Reilly [41]).

Corollary 2.9. *Let the hypotheses introduced in Sect.* 0.2 *be satisfied with* $\delta = 0$. *Let* $\theta \in \left(\max\left(0, \frac{1+\beta}{2} - \gamma\right), \frac{1}{2}\right)$, *put*

$$S_N(t) = N^{-1/2}\left\{\sum_{k=1}^{[Nt]} V_k - (Nt - [Nt])\, V_{[Nt]+1}\right\} \quad \textit{for } t \in [0,1],$$

and assume

$$\text{(2.9.C5)} \qquad N^{-1/2} \max_{n \leqq N} \left(\frac{N}{n}\right)^\theta \left\| \sum_{k=1}^{n} k^{-1/2}(T_k - T)\right\| \xrightarrow{P} 0 \quad \textit{as } N \to \infty.$$

Let $(H_N)_{N\in\mathbb{N}}$ *be a sequence in* $\mathcal{M}(\Omega, C^\theta_{\mathbb{B}}[0,1])$ *such that*

$$\text{(2.9.C6.a)} \qquad \lim_{\substack{t\to 0\\ t\neq 0}} t^{-\theta}\|H_N(t)\| = 0 \quad \textit{a.s.} \quad \textit{for all } N \in \mathbb{N},$$

$$\text{(2.9.C6.b)} \qquad \lim_{\varepsilon\to 0} \limsup_{N\to\infty} P^*\{\sup_{\substack{|s-t|\leqq\varepsilon\\ s,t\neq 0}} \|t^{-\theta} H_N(t) - s^{-\theta} H_N(s)\| \geqq \eta\} = 0 \quad \textit{for each } \eta \in \mathbb{R}^+$$

and

$$\text{(2.9.C6.c)} \qquad \|(S_N - H_N)(\cdot)\|_\theta \xrightarrow{P} 0 \quad \textit{as } N \to \infty.$$

Then - writing $\Lambda = \Gamma - \frac{1+\beta}{2} I$,

$$\text{(2.32)} \qquad \overline{H}_N(sN) = N^{1/2} H_N(s) \quad \textit{for } s \in [0,1], \quad N \in \mathbb{N}$$

and

$$\text{(2.33)} \qquad \hat{X}_N(t) = \overline{H}_N(t) - \Lambda \int_0^1 s^{\Lambda - I}\, \overline{H}_N(st)\, ds - t^{1/2}\left(\Gamma - \frac{\beta}{2} I\right)^{-1} T$$

$$\textit{for } N \in \mathbb{N}, \quad t \in [0, N] \text{ -}$$

one has

$$\text{(2.34)} \qquad N^{-1/2} \max_{0 \leqq t \leqq N} \|[t]^{(1+\beta)/2}\, U_{[t+1]} - \hat{X}_N(t)\| \xrightarrow{P^*} 0 \quad \textit{as } N \to \infty.$$

Proof. Adapting the argument used at the beginning of the proof of Corollary 2.7 and taking into account that $H_N \in \mathcal{M}(\Omega, C^\theta_{\mathbb{B}}[0,1])$, it becomes evident that

the integral in (2.33) is well-defined. Let $X_{n+1,N}$ be defined as $Y_{n+1,N}$ in (2.11) but with $Z_{k,N}=\bar{H}_N(k)$. Combining (2.9.C6.a)–(2.9.C6.c), we get

$$\lim_{r\to\infty}\limsup_{N\to\infty} P^*\{\|S_N(\cdot)\|_\theta \geqq r\}=0, \tag{2.35}$$

so that it follows from Theorem 2.2 that

$$N^{-1/2}\max_{0\leqq n\leqq N} n^{(1+\beta)/2}\|U_{n+1}-X_{n+1,N}\| \xrightarrow{P^*} 0 \quad \text{as } N\to\infty. \tag{2.36}$$

To complete the proof, it suffices to check that

$$N^{-1/2}\max_{0\leqq t\leqq N} \|[t]^{(1+\beta)/2} X_{[t+1],N}-\hat{X}_N(t)\| \xrightarrow{P} 0 \quad \text{as } N\to\infty. \tag{2.37}$$

Regarding the proof of (2.27), we see that it is enough to verify that

$$\max_{k\in\{1,\dots,N-1\}}\ \max_{k-1\leqq t\leqq k+1} \left(\frac{N}{k}\right)^\theta \left\|H_N\left(\frac{t}{N}\right)-H_N\left(\frac{k}{N}\right)\right\| \xrightarrow{P} 0 \quad \text{as } N\to\infty. \tag{2.38}$$

The maximum in (2.38) is $\leqq I_1(N)+I_2(N)$, where

$$I_1(N)=\max_{k\in\{1,\dots,N-1\}}\ \sup_{k-1<t\leqq k+1} \left\|\left(\frac{N}{t}\right)^\theta H_N\left(\frac{t}{N}\right)-\left(\frac{N}{k}\right)^\theta H_N\left(\frac{k}{N}\right)\right\|$$

and

$$I_2(N)=\max_{k\in\{1,\dots,N-1\}}\ \sup_{k-1<t\leqq k+1} \left\|\left(\frac{N}{t}\right)^\theta H_N\left(\frac{t}{N}\right)\left(1-\left(\frac{t}{k}\right)^\theta\right)\right\|.$$

Using (2.9.C6.b), we find that $I_1(N)\xrightarrow{P}0$ as $N\to\infty$. Finally, $I_2(N)\xrightarrow{P}0$ as $N\to\infty$ is implied by the estimate

$$\begin{aligned} I_2(N)\leqq &\max_{k\in\{1,\dots,M\}}\ \sup_{k-1<t\leqq k+1} \left\|\left(\frac{N}{t}\right)^\theta H_N\left(\frac{t}{N}\right)\right\| \\ &+\left(1-\left(\frac{M}{M+1}\right)^\theta\right)\max_{k\in\{1,\dots,N\}}\ \sup_{k-1<t\leqq k+1} \left\|\left(\frac{N}{t}\right)^\theta H_N\left(\frac{t}{N}\right)\right\| \end{aligned}$$

($M\in\{1,\dots,N-1\}$) together with (2.9.C6.a), (2.9.C6.b), (2.35) and (2.9.C6.c).

Corollary 2.10. *Let the hypotheses of Corollary 2.9 except those concerning the sequence (H_N) be fulfilled, and assume:*

(2.10.C6.a) $\mathbb{B}$ *is separable.*

(2.10.C6.b) *The V_n's are Borel measurable.*

(2.10.C6.c) *The sequence of random elements S_N in $(C^\theta_{\mathbb{B}}[0,1], \|\cdot\|_\theta)$ converges in distribution to a random element H in $C^\theta_{\mathbb{B}}[0,1]$.*

(2.10.C6.d) $\lim\limits_{\substack{t\to 0\\ t\neq 0}} t^{-\theta}\|H(t)\|=0$ *a.s.*

Then, without loss of generality, there exist copies H_N of H such that (2.34) *holds with $\hat{X}_N$ (and $\bar{H}_N$) defined as in* (2.33) (*and* 2.32)).

Proof. By (2.10.C6.a) and (2.10.C6.c), it follows from the Skorohod-Dudley theorem and Lemma 2.13 in [11] that, without loss of generality, there exists a sequence (H_N) of copies of H such that (2.9.C6.c) holds. The thus obtained sequence (H_N) also satisfies (2.9.C6.b) (due to the fact that any continuous function f: $[0,1]\to\mathbb{B}$ is uniformly continuous), so that the desired result is a consequence of Corollary 2.9.

Remark 2.11. In [55] (Lemma 2) Walk proved a related result, but with (2.10.C6.c) and (2.10.C6.d) replaced by

(2.11.C6.c) (a) $E\,\|S_n(1)\|=O(1)$,

(b) (2.10.C6.c) holds for $\theta=0$.

To elucidate the connection between these conditions, the following observation should be helpful:

Let $(V_n,\mathscr{F}_n)$ and $(\tilde{V}_n,\mathscr{F}_n)$ be two martingale difference sequences in $\mathbb{B}$ (cf., e.g., Métivier [36], p. 40) such that

$$\text{(i)}\qquad \lim_{N\to\infty} P\left(\bigcup_{k=N}^{\infty}\{V_k\neq\tilde{V}_k\}\right)=0,\qquad \text{(ii)}\ E\left\|n^{-1/2}\sum_{k=1}^{n}\tilde{V}_k\right\|=O(1)$$

and (iii) (2.10.C6.c) holds for $\theta=0$ with H being a $\mathbb{B}$-valued Brownian motion. Then, adapting the arguments used in the proof of Remark 2.5, it follows from Lemmas 3.1 and 3.2 that (2.10.C6.c) and (2.10.C6.d) are fulfilled for any $\theta\in(0,1/2)$.

For possible applications of Corollaries 2.9 and 2.10, the reader is referred to Walk [53], [55].

Let $\mathbb{D}=D[0,1]$ be the linear space of right-continuous functions f: $[0,1]\to\mathbb{R}$ with left limits, continuous at 1. With respect to the metric d_0 defined in Billingsley's [3] book (p. 109ff.), $(\mathbb{D},d_0)$ is a complete separable metric space and the topology induced by d_0 coincides with Skorohod's [48] $\mathbf{J}_1$-topology. Furnished with the supremum norm (denoted by $\|\cdot\|$), $(\mathbb{D},\|\cdot\|)$ is a nonseparable Banach space. For $\theta\in[0,1/2)$, let $C^{\theta}_{\mathbb{D}}[0,1]$ denote the linear space of all d_0-continuous functions f: $[0,1]\to\mathbb{D}$ such that $\sup_{t\in(0,1]} t^{-\theta}\|f(t)\|<\infty$. Equipped with the metric d^*_θ defined by $d^*_\theta(f,g)$ $=\sup_{t\in(0,1]} d_0(t^{-\theta}f(t),\,t^{-\theta}g(t))$, $(C^{\theta}_{\mathbb{D}}[0,1],\,d^*_\theta)$ is a complete separable metric space.

Corollary 2.12. *Let the hypotheses of Corollary* 2.9 *except those concerning the sequence (H_N) be fulfilled, and assume:*

(2.12.C6.a) $(\mathbb{B},\|\cdot\|)=(\mathbb{D},\|\cdot\|)$, *where $\|\cdot\|$ denotes the supremum norm on* $\mathbb{D}$.

(2.12.C6.b) *The V_n's are measurable with respect to the Borel σ-field induced by the $\mathbf{J}_1$-topology.*

(2.12.C6.c) *The sequence of random elements* S_N *in* $(C^\theta_{\mathbb{D}}[0,1], d^*_\theta)$ *converges in distribution to a random element* H *in* $C^\theta_{C^0_{\mathbb{R}}[0,1]}[0,1] \subset C^\theta_{\mathbb{D}}[0,1]$.

(2.12.C6.d) $\lim\limits_{\substack{t\to 0\\ t\neq 0}} t^{-\theta}\|H(t)\| = 0 \quad a.s.$

Then, without loss of generality, there exist copies H_N *of* H *such that* (2.34) *holds with* $\hat{X}_N$ *(and* $\bar{H}_N$*) defined as in* (2.33) *(and* (2.32)*).*

Proof. Compared to the proof of Corollary 2.10, some minor technical changes are necessary in order to establish (2.9.C6.c). Arguing as in the proof of Corollary 2.10, we can, without loss of generality, find copies H_N of H such that $d^*_\theta(S_N(\cdot), H_N(\cdot)) \xrightarrow{P} 0$ as $N\to\infty$. By (2.12.C6.c) and since d^*_θ-convergence of a sequence (f_n) of functions in $C^\theta_{\mathbb{D}}[0,1]$ to a function $f\in C^\theta_{C^0_{\mathbb{R}}[0,1]}[0,1]$ implies $\lim\limits_{n\to\infty}\|f_n - f\|_\theta = 0$ (cf. Walk [54], p. 448), this yields $\|(S_N - H_N)(\cdot)\|_\theta \xrightarrow{P} 0$ as $N\to\infty$.

Remark 2.13. Walk's [54] Theorem 4 is an immediate consequence of Theorem 1, Theorem 2 and Theorem 3 in [54] and Remark 2.11 and Corollary 2.12 of the present paper.

In the following corollary we consider a Robbins-Monro type algorithm in the case of weakly dependent (correlated) errors. The underlying model is closely related to that of Borodin [6].

Corollary 2.14. *Let* $\mathbb{H}$ *be a real separable Hilbert space (with norm* $\|\cdot\|$*) endowed with its Borel* σ*-field. Let* $M\colon \mathbb{H}\to\mathbb{H}$ *be a measurable function, Fréchet differentiable at* 0*, such that* $M(0)=0$ *and*

(2.14.C1) $\Gamma = M'(0)$ *satisfies* (0.2.C2) *for some* $\gamma > 1/2$.
Let $(G_n)_{n\in\mathbb{N}}$ *be a sequence in* $\mathcal{M}(\mathbb{H}\times\Omega, \mathbb{H})$*, and let* $(\mathcal{A}_n)_{n\in\mathbb{N}\cup\{0\}}$ *be a nondecreasing sequence of sub-*σ*-fields of* $\mathcal{A}$*. Suppose:*

(2.14.C0) $U_1\in\mathcal{M}(\Omega,\mathbb{H})$ *is* $\mathcal{A}_0$*-measurable and* $U_{n+1} = U_n - n^{-1}M(U_n) - n^{-1}G_n(U_n,\cdot)$ *for* $n\in\mathbb{N}$.

(2.14.C3) $U_n\to 0$ *a.s. as* $n\to\infty$.

(2.14.C5.a) *For all* $n\in\mathbb{N}$, $x\in\mathbb{H}$*, the mappings* $G_n(x,\cdot)$ *are* $\mathcal{A}_n$*-measurable.*

(2.14.C5.b) *For every bounded set* $B\subset\mathbb{H}$ *there is a sequence of square integrable functions* $\psi_{B,n}\in\mathcal{M}(\Omega,\mathbb{R}^+_0)$ *such that* $\sup\limits_{l\in\mathbb{N}} E\psi^2_{B,l} < \infty$ *and* $\|G_n(x,\cdot) - G_n(y,\cdot)\| \leqq \psi_{B,n}\|x-y\|$ *for all* $x, y\in B$, $n\in\mathbb{N}$.

(2.14.C5.c) $\sup\limits_{n\in\mathbb{N}} E\sup\limits_{x\in\mathbb{H}}\|G_n(x,\cdot)\|^r < \infty$ *for some* $r\in(2,\infty)$.

(2.14.C5.d) *There is a strictly increasing sequence* $(\lambda_k)_{k\in\mathbb{N}\cup\{0\}}$ *of integers such that* $\lambda_0 = 0$, $\lambda_{k+1} - \lambda_k = o(\lambda_{k+1}^{1/2})$ *and*

$$\lim_{n\to\infty}\lambda_n^{-1/2}\sum_{l=1}^{n}\sup_{x\in\mathbb{H}}\left\|\sum_{j=\lambda_{l-1}+1}^{\lambda_l} E(G_j(x,\cdot))\right\| = 0.$$

(2.14.C5.e) *There exist constants* $p, q \in \mathbb{R}^+$ *and a monotonically decreasing sequence* (ρ_k) *in* $\mathbb{R}^+$ *such that* $2<p<q<r$, $\sum_{k=1}^{\infty} \rho_k < \infty$ *and – with* $g_n(x) = E(G_n(x,\cdot))(x \in \mathbb{H})$ –

(i) $E\|E(G_n(\xi,\cdot) - g_n(\xi)|\mathscr{A}_{n-m})\|^p \leqq \rho_m^p (E\|G_n(\xi,\cdot)\|^q)^{p/q}$,

(ii) $E\|E(G_n(\xi,\cdot) - G_n(0,\cdot) - (g_n(\xi) - g_n(0))|\mathscr{A}_{n-m})\|^p \leqq \rho_m^p (E\|G_n(\xi,\cdot) - G_n(0,\cdot)\|^q)^{p/q}$ *for all* $(n,m) \in \mathbb{N}^2$ *with* $m<n$ *and any* $\mathscr{A}_{n-m}$*-measurable random variable* $\xi \in \mathscr{M}(\Omega, \mathbb{H})$.

Then – writing $Y_{n+1} = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^{\Gamma} k^{-1} G_k(0,\cdot)$ *for* $n \in \mathbb{N}$ *– one has*

$$N^{-1/2} \max_{n \leqq N} n \| U_{n+1} - Y_{n+1} \| \xrightarrow{P} 0 \quad \text{as } N \to \infty. \tag{2.39}$$

Proof. Since M is Fréchet differentiable at 0, M can be written in the form $M(x) = \tilde{\Gamma}(x)\, x$ $(x \in \mathbb{H})$, where $\tilde{\Gamma} \in \mathscr{M}(\mathbb{H}, \mathscr{L}(\mathbb{H}, \mathbb{H}))$ is continuous at 0. Thus

$$U_{n+1} = (I - n^{-1} \Gamma_n)\, U_n - n^{-1} V_n - n^{-3/2} T_n \tag{2.40}$$

with $\Gamma_n = \tilde{\Gamma}(U_n)$, $V_n = G_n(0,\cdot) - g_n(0)$ and $T_n = n^{1/2}\{G_n(U_n,\cdot) - G_n(0,\cdot) + g_n(0)\}$. We shall show that – for any $\theta \in (0, \frac{1}{2})$ – the just defined V_n's and T_n's obey conditions (2.2.C4) and (2.2.C5) (with $T=0$) of Theorem 2.2; taking $Z_{k,N} = V_k$ and perceiving that $\Gamma_n \to \Gamma$ a.s. as $n \to \infty$ (by (2.14.C3) and since $\tilde{\Gamma}$ is continuous at 0), (2.39) then follows from Theorem 2.2.

First we consider the V_n's. Combining the estimate

$$(E\|E(V_{n+k}|\mathscr{A}_n)\|^p)^{1/p} \leqq \rho_k (E\|G_{n+k}(0,\cdot)\|^q)^{1/q} \qquad (k, n \in \mathbb{N}) \tag{2.41}$$

(implied by (2.14.C5.e)(i)), (2.14.C5.c), the fact that $\sum_{j=1}^{\infty} \rho_j < \infty$, Hölder's and Minkowski's inequality, we infer that – for each $n \in \mathbb{N}$ – the sequence $(R_n^{(m)})_{m \in \mathbb{N}}$ defined by $R_n^{(m)} = \sum_{k=1}^{m} E(V_{n+k}|\mathscr{A}_n)$ converges in p-th mean and a.s. to a random variable R_n as $m \to \infty$. Taking into account that – for each $n \in \mathbb{N}$ – V_n is $\mathscr{A}_n$-measurable (by (2.14.C5.a)), we obtain the representation

$$V_n = \zeta_n - R_n + R_{n-1}, \tag{2.42}$$

where (ζ_n) is a martingale difference sequence (cf. Hall and Heyde [20], p. 127ff., or Philipp and Stout [43], p. 7f.). It is clear from (2.14.C5.c) and the definition of the R_n's that

$$\sup_{n \in \mathbb{N}} E\|R_n\|^p < \infty \quad \text{and} \quad \sup_{n \in \mathbb{N}} E\|V_n\|^2 < \infty \tag{2.43}$$

(by Hölder's and Minkowski's inequality). This entails that the second absolute moments of the sequence (ζ_n) are uniformly bounded, i.e., $E\left\|\sum_{k=1}^{n} \zeta_k\right\| = O(n^{1/2})$.

Therefore, using Lemma 3.1,

$$\lim_{r\to\infty}\limsup_{N\to\infty} P\left\{N^{-1/2}\max_{n\leqq N}\left(\frac{N}{n}\right)^{\theta}\left\|\sum_{k=1}^{n}\zeta_k\right\|\geqq r\right\}=0. \tag{2.44}$$

Since $n^{-1/2}R_n\to 0$ a.s. as $n\to\infty$ (by (2.43), the Chebyshev-Markov inequality and the Borel-Cantelli lemma), (2.44) remains true with V_k in place of ζ_k.

We now turn to the T_n's. Because of the properties of the sequence (ρ_k), we can find a sequence $(\mu(k))_{k\in\mathbb{N}}$ in $\mathbb{N}$ such that

$$1<\mu(k)<k \text{ for all } k>2, \quad \mu(k)=o(k^{1/2}) \text{ and } \rho_{\mu(k)}=o(k^{-1/2}). \tag{2.45}$$

Let $\varepsilon\in(0,1/2)$ be given. By (2.14.C3) there is a constant $K(\varepsilon)$ in $\mathbb{R}^+$ such that the set $A(\varepsilon)=\{\sup_{n\in\mathbb{N}}\|U_n\|\geqq K(\varepsilon)\}$ has probability $\leqq\varepsilon$. Let $\hat{T}_n$ be defined as T_n but with $U_n\chi(\{\|U_n\|\leqq K(\varepsilon)\})$ instead of U_n. Then $P\left(\bigcup_{n=1}^{\infty}\{T_n\neq\hat{T}_n\}\right)\leqq\varepsilon$. Since $\varepsilon\in(0,1/2)$ is arbitrary, it suffices to show that

$$N^{-1/2}\max_{3\leqq n\leqq N}\left(\frac{N}{n}\right)^{\theta}\left\|\sum_{k=3}^{n}k^{-1/2}\hat{T}_k\right\|\xrightarrow{P}0 \quad \text{as } N\to\infty. \tag{2.46}$$

In the rest of the proof we shall always assume that $n\geqq 3$. To prove (2.46), we start from the decomposition

$$\hat{T}_n=n^{1/2}\sum_{j=1}^{4}\hat{T}_n^{(j)}, \tag{2.47}$$

where

$$\hat{T}_n^{(1)}=\hat{T}_n^*-E(\hat{T}_n^*|\mathscr{A}_{n-\mu(n)}), \quad \hat{T}_n^{(2)}=E(\hat{T}_n^*|\mathscr{A}_{n-\mu(n)}),$$
$$\hat{T}_n^{(3)}=g_n(\hat{U}_{n-\mu(n)}) \quad \text{and} \quad \hat{T}_n^{(4)}=G_n(\hat{U}_n,\cdot)-G_n(\hat{U}_{n-\mu(n)},\cdot)$$

with

$$\hat{U}_n=U_n\chi(\{\|U_n\|\leqq K(\varepsilon)\}) \quad \text{and}$$
$$\hat{T}_n^*=G_n(\hat{U}_{n-\mu(n)},\cdot)-G_n(0,\cdot)-(g_n(\hat{U}_{n-\mu(n)})-g_n(0)).$$

Put $\nu(n)=\max\{k\in\mathbb{N}:\ n+k-\mu(n+k)<n\}$, and let $\hat{R}_n$ be defined by $\hat{R}_n=\sum_{k=1}^{\nu(n)}E(\hat{T}_{n+k}^{(1)}|\mathscr{A}_n)$. Then, noting that $E(\hat{T}_{n+k}^{(1)}|\mathscr{A}_n)=0$ a.s. for $k>\nu(n)$, the following analogue of (2.42) holds true:

$$\hat{T}_n^{(1)}=\hat{\zeta}_n-\hat{R}_n+\hat{R}_{n-1}, \tag{2.48}$$

where $(\hat{\zeta}_k)_{k\geqq 3}$ is a martingale difference sequence. Let

$$\Delta_n=\|G_n(\hat{U}_{n-\mu(n)},\cdot)-G_n(0,\cdot)\|.$$

Then – using the inequality $E\|E(X|\mathscr{F})\|^2\leqq E\|X\|^2$ (valid for all random variables $X\in\mathscr{M}(\Omega,\mathbb{H})$ with $E\|X\|^2<\infty$ and any sub-σ-field $\mathscr{F}$ of $\mathscr{A}$), (2.14.C5.e)(ii), Hölder's and Minkowski's inequality –

$$(E\|\hat{T}_n^{(1)}\|^2)^{1/2}\leqq 2(E\|\hat{T}_n^*\|^2)^{1/2}\leqq 4(E\Delta_n^2)^{1/2}+2\rho_{\mu(n)}(E\Delta_n^q)^{1/q} \tag{2.49}$$

and

$$(2.50)\qquad (E\|\hat{R}_n\|^p)^{1/p} \leqq \sum_{k=1}^{v(n)} (E\|E(\hat{T}_{n+k}^{(1)}|\mathscr{A}_n)\|^p)^{1/p}$$
$$\leqq 2\sum_{k=1}^{\infty} \rho_k (E\Delta_{n+k}^q)^{1/q}.$$

Combining (2.14.C3), the fact that (by (2.14.C5.b)) the mappings $G_n(\cdot,\omega)$ are continuous for each $\omega\in\Omega$, (2.14.C5.e) and Satz 1.14.7 in [17], it follows that

$$(2.51)\qquad \Delta_n\to 0 \text{ a.s., } E\Delta_n^q\to 0 \text{ and } E\Delta_n^2\to 0 \text{ as } n\to\infty.$$

(Recall that $2<p<q<r$.) Hence

$$(2.52)\qquad E\|\hat{T}_n^{(1)}\|^2\to 0 \text{ and } E\|\hat{R}_n\|^p\to 0 \text{ as } n\to\infty,$$

which implies that $E\|\hat{\zeta}_n\|^2=o(1)$ and thus $E\left\|\sum_{k=1}^{n}\hat{\zeta}_k\right\|=o(n^{1/2})$. Applying Lemma 3.1 $\left(\text{with } S_k=\left\|\sum_{j=1}^{k}\hat{\zeta}_j\right\|\right)$ and taking into account that $n^{-1/2}\hat{R}_n\to 0$ a.s. as $n\to\infty$ (by (2.52)), we see that (2.46) holds with $\hat{T}_k^{(1)}$ in place of $k^{-1/2}\hat{T}_k$.

Let $A_n=g_n(\hat{U}_{\lambda_k+1-\mu(\lambda_k+1)})$ for $k>2$ and $\lambda_k<n\leqq\lambda_{k+1}$, $A_n=\hat{T}_n^{(3)}$ for $3\leqq n<\lambda_3$, and $B_n=\hat{T}_n^{(3)}-A_n$ for $n\geqq 3$. Since $U_n\to 0$ a.s. as $n\to\infty$ and $\tilde{\Gamma}$ is continuous at 0, we can find a constant $L(\varepsilon)$ in $\mathbb{R}^+$ such that the set $\Omega_\varepsilon=\Omega\setminus A(\varepsilon)\setminus\{\sup_{n\in\mathbb{N}}(\|\Gamma_n U_n\| - L(\varepsilon)\|U_n\|)\geqq 0\}$ has probability $\geqq 1-2\varepsilon$. $\varepsilon\in(0,1/2)$ was arbitrarily prescribed; thus, in view of Remark 2.3, the proof would be finished if we knew that

$$(2.53)\quad \text{(a)}\quad E\|\hat{T}_n^{(2)}\|=o(n^{-1/2}),\qquad \text{(b)}\quad \left\|\sum_{k=3}^{n}A_k\right\|=o(n^{1/2}) \text{ a.s. on } \Omega_\varepsilon,$$
$$\text{(c)}\quad E\|B_n\|\,\chi(\Omega_\varepsilon)=o(n^{1/2}),\quad \text{(d)}\quad E\|\hat{T}_n^{(4)}\|\,\chi(\Omega_\varepsilon)=o(n^{-1/2}).$$

We have

$$(2.53)\quad \text{(a)}\quad E\|\hat{T}_n^{(2)}\|\leqq(E\|\hat{T}_n^{(2)}\|^p)^{1/p}\leqq\rho_{\mu(n)}(E\Delta_n^q)^{1/q}=o(n^{-1/2})$$
(by (2.14.C5.e)(ii), (2.45) and (2.51)).

Writing $B(\varepsilon)=\{x\in\mathbb{H}: \|x\|\leqq K(\varepsilon)\}$, it follows from (2.14.C5.b) that

$$(2.54)\qquad \|g_n(x)-g_n(y)\|\leqq E\|G_n(x,\cdot)-G_n(y,\cdot)\|$$
$$\leqq\sup_{n\in\mathbb{N}}(E\psi_{B(\varepsilon),n}^2)^{1/2}\|x-y\|$$

for all $x,y\in B(\varepsilon)$. Combining (2.54) and (2.14.C3) yields: $\|A_n\|=o(1)$ a.s. on Ω_ε. Letting $n\geqq\lambda_3$ and using the notation $\kappa(n)=\max\{k\in\mathbb{N}: \lambda_k\leqq n\}$, this together with (2.14.C5.d) leads to

$$(2.53)\quad \text{(b)}\quad \left\|\sum_{k=\lambda_3}^{n}A_k\right\|\leqq\sum_{j=3}^{\kappa(n)}\left\|\sum_{k=\lambda_j+1}^{\lambda_{j+1}}A_k\right\|+\sum_{k=\lambda_{\kappa(n)}+1}^{\lambda_{\kappa(n)+1}}\|A_k\|=o(n^{1/2})$$

a.s. on Ω_ε.

By (2.14.C5.c) and the definition of Ω_ε, we have

$$\begin{aligned}(E\|\hat{U}_{n+1}-\hat{U}_n\|^2\chi(\Omega_\varepsilon))^{1/2} &\leqq n^{-1}((E\|\Gamma_n\hat{U}_n\|^2\chi(\Omega_\varepsilon))^{1/2}\\ &\quad+(E\|G_n(\hat{U}_n,\cdot)\|^2\chi(\Omega_\varepsilon))^{1/2})\\ &=O(n^{-1}),\end{aligned}$$

which implies that

$$\sup_{1\leqq k\leqq\gamma(n)}(E\|U_n-U_{n-k}\|^2\chi(\Omega_\varepsilon))^{1/2}=o(n^{-1/2}) \tag{2.55}$$

for any sequence $(\gamma(n))_{n\in\mathbb{N}}$ in $\mathbb{N}$ with $\gamma(n)=o(n^{1/2})$. Consequently, using (2.54),

$$E\|B_n\|\chi(\Omega_\varepsilon)=o(n^{-1/2}). \tag{2.53 (c)}$$

Finally, on account of (2.14.C5.b),

$$\begin{aligned}E\|\hat{T}_n^{(4)}\|\chi(\Omega_\varepsilon) &\leqq \sup_{n\in\mathbb{N}}(E\psi^2_{B(\varepsilon),n})^{1/2}(E\|\hat{U}_n-\hat{U}_{n-\mu(n)}\|^2\chi(\Omega_\varepsilon))^{1/2}\\ &=o(n^{-1/2})\quad\text{(by (2.45) and (2.55))}.\end{aligned} \tag{2.53 (d)}$$

Remark 2.15. Sufficient conditions for (2.14.C3) (i.e., for $U_n\to 0$ a.s. as $n\to\infty$) to be satisfied have been derived (e.g.) by Borodin [6] and Watanabe [56].

Corollary 2.16. *Let the hypotheses of Corollary* 2.14 *be fulfilled. Proceeding from* $V_n=G_n(0,\cdot)-g_n(0)$ $(n\in\mathbb{N})$, *let* S_N *be defined as in Corollary* 2.9. *Also assume:*

(2.16.C6.a) $\mathbb{H}=\mathbb{R}^d$ (*for some* $d\in\mathbb{N}$).

(2.16.C6.b) *There is a symmetric nonnegative definite* $d\times d$ *matrix* S *such that, for all* $u,v,w\in[0,1]$ *with* $u<v<w$,

$$\lim_{N\to\infty}E\left\|N^{-1}E\left(\left(\sum_{k=[Nv]}^{[Nw]}V_k\right)\left(\sum_{k=[Nv]}^{[Nw]}V_k\right)^T|\mathscr{A}_{[Nu]}\right)-(w-v)S\right\|=0.$$

(*The elements of* $\mathbb{R}^d$ *are considered as column vectors;* x^T *denotes the transpose of a vector* $x\in\mathbb{R}^d$.) *Then, without loss of generality, there exist a Brownian motion* W *in* $\mathbb{R}^d$ *with* $W(0)=0$ *and* $\mathrm{Cov}(W(1))=S$ *and a sequence of copies* H_N *of* W *such that* (2.34) *holds with* $\hat{X}_N$ *and* $\overline{H}_N$ *defined by* (2.33) (*case* $\beta=1$ *and* $T=0$) *and* (2.32).

Proof. We shall first demonstrate that – for any $\theta\in[0,1/2)$ – the sequence of random elements S_N in $(C^\theta_{\mathbb{R}^d}[0,1],\|\cdot\|_\theta)$ (cf. Corollary 2.9) converges in distribution to a Brownian motion W in $\mathbb{R}^d$ with $W(0)=0$ and $\mathrm{Cov}(W(1))=S$. To this end, we start from the observation that the special case $\theta=0$ and $d=1$ of this assertion is a consequence of McLeish's [35] Theorem 2.4; indeed, using (2.14.C5.c), (2.14.C5.e)(i) and (2.16.C6.b), it is easily verified that the array $(X_{n,i})$ defined by $X_{n,i}=n^{-1/2}V_i$ $(i\in\{1,\ldots,n\},\ n\in\mathbb{N})$ satisfies the conditions of McLeish's theorem (see also Remark 2.7 in [35]). By virtue of the Cramèr-Wold device and the fact that tightness can be proved componentwise, this result carries over to arbitrary $d\in\mathbb{N}$. The argument for proving that the assumption $\theta=0$ can also be removed depends on the representation (2.42). In the proof of Corollary 2.14 it was shown that $n^{-1/2}R_n\to 0$ a.s., which entails

that the partial sums of the sequence (V_k) satisfy a functional CLT in $(C^\theta_{\mathbb{R}^d}[0,1], \|\cdot\|_\theta)$ (for any $\theta\in[0,1/2)$) with limit process W iff the partial sums of (ζ_k) do, and that $E\left\|\sum_{k=1}^n \zeta_k\right\| = O(n^{1/2})$. According to Remark 2.11 this implies that, with W in place of H, (2.10.C6.c) holds true for any $\theta\in(0,1/2)$, which concludes the first part of the proof. The assertion of the corollary now follows by combining Corollary 2.10 (as mentioned in Remark 2.11, (2.10.C6.d) is fulfilled with W in place of H), Theorem 2.2 and Corollary 2.14.

Remark 2.17. Corollary 2.16 improves the discrete case of Borodin's [6] Theorem 5.1 in both respects,

(a) the conclusion is stronger (Borodin only proved weak convergence of the finite dimensional distributions, while Corollary 2.16 is equivalent to a weak invariance principle (cf. Remark 2.8))

and

(b) the hypotheses are weaker.

Since (b) is not obvious at the first glance, we now add some facts which are essential ingredients for checking (b), the remaining details being left to the reader:

(i) Presumably due to a misprint, in condition 1) of Borodin's theorem "in probability" should be replaced by "with probability 1"; otherwise it would not be possible to reduce the investigation of the process $(X(t))$ to that of $(\hat{X}(t))$ (see also Nevel'son and Has'minskiĭ [38], p. 142f.).

(ii) Let $(\mathscr{G}_{m,n})_{m,n\in\mathbb{N}\cup\{0\},m\leq n}$ be an increasing family of sub-σ-fields of $\mathscr{A}$ (i.e., $\mathscr{G}_{m,n}\subset\mathscr{G}_{m',n'}$ if $m'\leqq m$ and $n'\geqq n$) satisfying a strong mixing condition (cf. Borodin [6], p. 36) with mixing rate $(\alpha_n)_{n\in\mathbb{N}}$, and let $(X_n)_{n\in\mathbb{N}}$ be a sequence of $\mathbb{R}^d$-valued random variables, centered at expectations and having uniformly bounded r-th absolute moments for some $r\in(2,\infty)$, such that, for each $n\in\mathbb{N}$, X_n is $\mathscr{G}_{n,n}$-measurable. Suppose:

(*) There is a symmetric nonnegative definite $d\times d$ matrix S such that

$$\lim_{N\to\infty} N^{-1} E\left(\left(\sum_{k=1}^{[Nt]} X_k\right)\left(\sum_{k=1}^{[Nt]} X_k\right)^T\right) = tS \quad \text{for all } t\in[0,1].$$

$$(**) \qquad \sum_{n=1}^{\infty} \alpha_n^{(1-\varepsilon)(r-2)/r} < \infty \quad \text{for some } \varepsilon\in(0,1).$$

Then, for all $u,v,w\in[0,1]$ with $u<v<w$,

$$(***) \qquad \lim_{N\to\infty} E\left\|N^{-1} E\left(\left(\sum_{k=[Nv]}^{[Nw]} X_k\right)\left(\sum_{k=[Nv]}^{[Nw]} X_k\right)^T \middle| \mathscr{G}_{0,[Nu]}\right) - (w-v)S\right\| = 0.$$

Apart from minor modifications, the proof of (***) can be adapted from McLeish [35], p. 620 (see also Remark 2.7 in [35]).

(iii) Arguing as Dehling [9] in the proof of his Lemma 3.8 (a prior version of which can be found in McLeish [33], Lemma 3.5), it follows that under the

hypotheses of Borodin's Lemma 2.1 (and using the same notation as there)

$$E|E(H(\xi,\cdot)|\mathcal{N}_0^s)-h(\xi)|^\tau \leqq c\,\beta(t-s)^{(\delta-\tau)/\delta}(E\sup_{x\in\mathbb{R}}|H(x,\cdot)|^\delta)^{\tau/\delta}$$

for any $\tau\in(1,\delta)$, where $c\in\mathbb{R}^+$ depends only on τ and δ.

The following theorem is closely related to Mark's [32] Satz 2.1.

Theorem 2.18. *Let the hypotheses introduced in Sect. 0.2 be satisfied with $\delta=0$. Suppose*

(2.18.C4) $$\limsup_{N\to\infty}(N\,\mathrm{L}_2 N)^{-1/2}\left\|\sum_{k=1}^N V_k\right\|<\infty \quad a.s.,$$

(2.18.C5) $$\lim_{N\to\infty}(N\,\mathrm{L}_2 N)^{-1/2}\left\|\sum_{k=1}^N k^{-1/2}T_k\right\|=0 \quad a.s.$$

Let $(Z_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{M}(\Omega,\mathbb{B})$ such that

(2.18.C6) $$\lim_{N\to\infty}(N\,\mathrm{L}_2 N)^{-1/2}\left\|\sum_{k=1}^N (V_k-Z_k)\right\|=0 \quad a.s.$$

Then – writing $Y_{n+1}=\sum_{k=1}^n\left(\frac{k}{n}\right)^\Gamma k^{-\frac{1+\beta}{2}}Z_k$ for $n\in\mathbb{N}$ – one has

(2.56) $$\lim_{N\to\infty}N^{\beta/2}(\mathrm{L}_2 N)^{-1/2}\|U_{N+1}-Y_{N+1}\|=0 \quad a.s.$$

Proof. The proof is similar to that of Theorem 2.2. Let $\omega\in\Omega$ be arbitrarily fixed, and choose $\gamma_k=\Gamma_k(\omega)$ $(k\in\mathbb{N})$. We have

$$U_{n+1}(\omega)=\sum_{j=1}^3 U_{n+1}^{(j)}(\omega),$$

where the $U_{n+1}^{(j)}(\omega)$'s are defined as in (2.13). (2.56) will be established by showing that the mappings $U_{n+1}^{(j)}$ defined by (2.13) satisfy

(2.57) $$\lim_{N\to\infty}N^{\beta/2}(\mathrm{L}_2 N)^{-1/2}\|U_{N+1}^{(1)}\|=0 \quad \text{a.s.},$$

(2.58) $$\lim_{N\to\infty}N^{\beta/2}(\mathrm{L}_2 N)^{-1/2}\|U_{N+1}^{(2)}-Y_{N+1}\|=0 \quad \text{a.s.},$$

(2.59) $$\lim_{N\to\infty}N^{\beta/2}(\mathrm{L}_2 N)^{-1/2}\|U_{N+1}^{(3)}\|=0 \quad \text{a.s.}$$

(2.57) is an immediate consequence of (0.2.C3) and Lemma 3.3 below. To prove (2.58), we apply Proposition 2.1 with

$$J=\{N\},\quad v_{k,N}=V_k(\omega),\quad z_{k,N}=Z_k(\omega),\quad \lambda(m)=\left(\frac{m}{N}\right)^{1/2},$$

$$u_{n,N}^{(0)}=U_{n+1}^{(2)}(\omega),\quad y_{n,N}=Y_{n+1}(\omega),\quad \tau=\rho=(1+\beta)/2,\quad \delta=0,\quad \theta=1/2.$$

Using the same notation as there, we find that – for $\omega\in\Omega\setminus M_0$ (with M_0 being defined in Sect. 0.2) –

(2.60) $$C_N(J,A',\lambda)=o(1)\ \ (\text{as } N\to\infty)\quad\text{and}\quad C_N(J,A'',\lambda)=O(1)$$

(since $\kappa>0$ and $a_n=1$ for all n). Combining (2.60),

$$\Delta_N(J, v^{(N)}, \lambda)=\max_{n\leqq N}\left(\frac{N}{n}\right)^{1/2}\left\|\sum_{k=1}^{n} V_k(\omega)\right\|, \tag{2.61}$$

$$\Delta_N(J, v^{(N)}-z^{(N)}, \lambda)=\max_{n\leqq N}\left(\frac{N}{n}\right)^{1/2}\left\|\sum_{k=1}^{n} (V_k(\omega)-Z_k(\omega))\right\|, \tag{2.62}$$

(2.18.C4) and (2.18.C6), (2.58) now follows from Proposition 2.1 (a). The proof of (2.59) proceeds analogously (take $v_{k,N}=k^{-1/2}\,T_k(\omega)$ and $z_{k,N}=0$).

Remark 2.19. In most applications, the partial sums of the sequence (Z_k) will be connected to a Brownian motion W (with $W(0)=0$), taking values in a separable closed linear subspace $\mathbb{B}_1$ of $\mathbb{B}$, via $W(n)=\sum_{k=1}^{n} Z_k$ (for all $n\in\mathbb{N}$). Then, in consequence of Fernique's [16] theorem and the Borel-Cantelli lemma, (2.18.C6) is equivalent to

$$\lim_{\substack{t\to\infty\\ t>0}} (t\,\mathrm{L}_2\,t)^{-1/2}\,\|W(t)-\sum_{k\leqq t} V_k\|=0 \quad \text{a.s.} \tag{2.19.C6}$$

Starting from Strassen's [49], [50] pioneering work, almost sure invariance principles of the form (2.19.C6) have been obtained under various conditions, among others, by Philipp and Stout [43], Hall and Heyde [19], Philipp [44], Kuelbs and Philipp [27], Morrow and Philipp [37] and Dudley and Philipp [11].

Corollary 2.20. *Let the hypotheses of Theorem* 2.18 *be fulfilled with* (2.18.C6) *replaced by* (2.19.C6), *where W is defined as in Remark* 2.19. *Then – writing* $\Lambda=\Gamma-\frac{1+\beta}{2}I$ and $\hat{Y}(t)=W(t)-\Lambda\int_0^1 s^{\Lambda-I}\,W(st)\,ds$ *for* $t\in\mathbb{R}_0^+$ *– one has*

$$\lim_{\substack{t\to\infty\\ t>0}} (t\,\mathrm{L}_2\,t)^{-1/2}\,\|[t]^{(1+\beta)/2}\,U_{[t+1]}-\hat{Y}(t)\|=0 \quad \text{a.s.} \tag{2.63}$$

Proof. The proof is similar to that of Corollary 2.7. The (a.s.) existence of the integral occurring in the definition of $\hat{Y}(t)$ has already been verified there. On account of Theorem 2.18, it remains only to show that

$$\lim_{\substack{t\to\infty\\ t>0}} (t\,\mathrm{L}_2\,t)^{-1/2}\,\|[t]^{(1+\beta)/2}\,Y_{[t+1]}-\hat{Y}(t)\|=0 \quad \text{a.s.} \tag{2.64}$$

Adapting the proof of (2.27), we see that the following stronger statement holds true:

$$\limsup_{\substack{t\to\infty\\ t>0}} t^{-\theta}\,\|[t]^{(1+\beta)/2}\,Y_{[t+1]}-\hat{Y}(t)\|<\infty \quad \text{a.s.} \tag{2.65}$$

for any $\theta>\max\left(0,\frac{1+\beta}{2}-\gamma\right)$.

Remark 2.21. Combining Corollary 2.20 and Theorem 1 of Kuelbs and LePage [26], it is possible to characterize the set of almost sure cluster points in the Banach space $(C_{\mathbb{B}}^{(0)}[0,1], \|\cdot\|_0)$ (defined before Corollary 2.9) of the sequence $(\hat{U}_N^*(\cdot))_{N\in\mathbb{N}}$ defined by

$$\hat{U}_N^*(t) = (2N\,\mathrm{L}_2 N)^{-1/2}\{R_{[Nt]} + (Nt - [Nt])(R_{[Nt]+1} - R_{[Nt]})$$

$(t\in[0,1])$ with $R_n = n^{(1+\beta)/2}\,U_{n+1}$. (For further details, the reader is referred to Mark [32], p. 7ff.) The thus obtained functional LIL comprises the more classical LIL type results (cf., e.g., Kuelbs and LePage [26], Corollaries 1–4).

The following theorem can be useful to describe the almost sure behaviour of the sequence (U_N) in cases where an almost sure invariance principle of the form (2.19.C6) does not hold (or is not available).

Theorem 2.22. *Let the hypotheses introduced in Sect.* 0.2 *be satisfied with* $\delta = 0$. *Suppose*

$$\limsup_{N\to\infty} (N\,\mathrm{L}_2 N)^{-1/2}\left(\left\|\sum_{k=1}^N V_k\right\| + \left\|\sum_{k=1}^N k^{-1/2}\,T_k\right\|\right) < \infty \quad a.s. \tag{2.22.C4}$$

Then

$$\limsup_{N\to\infty} N^{\beta/2}(\mathrm{L}_2 N)^{-1/2}\,\|U_{N+1}\| < \infty \quad a.s. \tag{2.66}$$

Proof. Apart from obvious modifications, the proof of Theorem 2.18 also applies to this situation.

Theorem 2.23. *Let the hypotheses introduced in Sect.* 0.2 *be satisfied for some* $\delta\in(0,1)$. *Let* $\varepsilon\in[0,\min(\delta,\gamma-\beta/2))$, $\eta\in(0,\min(1/2,\gamma-\beta/2-\varepsilon))$, *and assume*

$$\left\|\sum_{k=1}^N V_k\right\| = O(N^{1/2+\varepsilon}) \quad a.s., \tag{2.23.C4}$$

$$\left\|\sum_{k=1}^N k^{-1/2}(T_k - T)\right\| = O(N^{1/2-\eta}) \quad a.s. \tag{2.23.C5}$$

Let $(Z_n)_{n\in\mathbb{N}}$ *be a sequence in* $\mathscr{M}(\Omega,\mathbb{B})$ *such that*

$$\left\|\sum_{k=1}^N (V_k - Z_k)\right\| = O(N^{1/2-\eta}) \quad a.s. \tag{2.23.C6}$$

Then – writing

$$Y_{n+1} = \sum_{k=1}^n \left(\frac{k}{n}\right)^{\Gamma} k^{-\frac{1+\beta}{2}} Z_k - n^{-\beta/2}\left(\Gamma - \frac{\beta}{2} I\right)^{-1} T$$

for $n\in\mathbb{N}$ *– one has*

$$\lim_{N\to\infty} N^{\beta/2+\sigma}\,\|U_{N+1} - Y_{N+1}\| = 0 \quad a.s. \tag{2.67}$$

for any $\sigma\in[0,\min(\delta-\varepsilon,\eta,\gamma-\beta/2-\varepsilon-\eta))$.

Proof. The proof is similar to that of Theorem 2.2. Let $\omega\in\Omega$ be arbitrarily fixed, and choose $\gamma_k=\Gamma_k(\omega)$ ($k\in\mathbb{N}$). We have

$$U_{n+1}(\omega)=\sum_{j=1}^{3} U_{n+1}^{(j)}(\omega) \quad \text{and} \quad Y_{n+1}(\omega)=\sum_{j=1}^{3} Y_{n+1}^{(j)}(\omega),$$

where the $U_{n+1}^{(j)}(\omega)$'s are defined as in (2.13) and the $Y_{n+1}^{(j)}(\omega)$'s are defined analogously to the $Y_{n+1,N}^{(j)}(\omega)$'s in (2.14) (with Z_k instead of $Z_{k,N}$). (2.67) will be established by showing that the thus defined mappings $U_{n+1}^{(j)}$ and $Y_{n+1}^{(j)}$ satisfy

$$\lim_{N\to\infty} N^{\beta/2+\sigma}\|U_{N+1}^{(1)}\|=0 \quad \text{a.s.,} \tag{2.68}$$

$$\lim_{N\to\infty} N^{\beta/2+\sigma}\|U_{N+1}^{(j+1)}-Y_{N+1}^{(j)}\|=0 \quad \text{a.s. for } j\in\{1,2\}, \tag{2.69}$$

$$\lim_{N\to\infty} N^{\beta/2+\sigma}\|Y_{N+1}^{(3)}\|=0 \quad \text{a.s.} \tag{2.70}$$

(2.68) is an immediate consequence of (0.2.C3) and Lemma 3.3 below. The proof of (2.69) will be carried out only in the case $j=1$. (The case $j=2$ can be treated analogously.) Applying Proposition 2.1 with

$$J=\{N\}, \quad v_{k,N}=V_k(\omega), \quad z_{k,N}=Z_k(\omega), \quad \lambda(m)=\left(\frac{m}{N}\right)^{1/2-\eta},$$

$$u_{n,N}^{(0)}=U_{n+1}^{(2)}(\omega), \quad y_{n,N}=Y_{n+1}^{(1)}(\omega), \quad \tau=\rho=(1+\beta)/2,$$

$$\delta\in(0,1), \quad \theta=1/2-\eta$$

and using the same notation as there, we find that - for $\omega\in\Omega\setminus M_0$ (with M_0 being defined in Sect. 0.2) -

$$C_N(J,A',\lambda)=O(N^{-a}) \quad \text{for any } a\in(0,\min(\delta,\gamma-\beta/2-\eta)) \tag{2.71}$$

and

$$C_N(J,A'',\lambda)=O(1). \tag{2.72}$$

Combining (2.71), (2.72),

$$\Delta_N(J,v^{(N)},\lambda)=\max_{n\leq N}\left(\frac{N}{n}\right)^{1/2-\eta}\left\|\sum_{k=1}^{n} V_k(\omega)\right\|, \tag{2.73}$$

$$\Delta_N(J,v^{(N)}-z^{(N)},\lambda)=\max_{n\leq N}\left(\frac{N}{n}\right)^{1/2-\eta}\left\|\sum_{k=1}^{n} (V_k(\omega)-Z_k(\omega))\right\|, \tag{2.74}$$

(2.23.C4) and (2.23.C6), the case $j=1$ of (2.69) is implied by Proposition 2.1 (a). (2.70) can be rewritten in the form

$$\lim_{N\to\infty} N^{\sigma}\left\|\frac{1}{N}\sum_{k=1}^{N}\left(\frac{k}{N}\right)^{\Gamma-\left(1+\frac{\beta}{2}\right)I}-\left(\Gamma-\frac{\beta}{2}I\right)^{-1}\right\|=0. \tag{2.75}$$

Setting $\Lambda=\Gamma-\left(1+\frac{\beta}{2}\right)I$, (2.75) follows from the equality $\int_0^1 t^\Lambda\,dt=\left(\Gamma-\frac{\beta}{2}I\right)^{-1}$ (already mentioned in the proof of Theorem 2.2) and the estimate

$$(2.76)\quad \left\|\frac{1}{N}\sum_{k=1}^{N}\left(\frac{k}{N}\right)^\Lambda-\int_0^1 t^\Lambda\,dt\right\|$$

$$\leqq\frac{1}{N}\|N^{-\Lambda}\|+\int_0^{1/N}\|t^\Lambda\|\,dt+\sum_{k=2}^{N}\int_{(k-1)/N}^{k/N}\int_{(k-1)/N}^{k/N}\|\Lambda s^{\Lambda-I}\|\,ds\,dt$$

(by the mean value theorem)

$$=O\left(\frac{1}{N}\int_{1/N}^{1}s^{\gamma-\beta/2-2}\,ds\right)\quad\text{(by (0.2.C2))}$$

$$=o(N^{-\sigma})\quad\text{as } N\to\infty.$$

Corollary 2.24. *Let W be a Brownian motion on $(\Omega,\mathscr{A},P)$ with $W(0)=0$, taking values in a separable closed linear subspace $\mathbb{B}_1$ of $\mathbb{B}$, and let the hypotheses of Theorem 2.23 be satisfied with (2.23.C6) replaced by*

$$(2.24.\text{C6})\qquad \|W(t)-\sum_{k\leqq t}V_k\|=O(t^{1/2-\eta})\quad a.s.\ (for\ t\geqq 1).$$

Then – writing $\Lambda=\Gamma-\frac{1+\beta}{2}I$ and

$$(2.77)\qquad \hat{Y}(t)=W(t)-\Lambda\int_0^1 s^{\Lambda-I}W(st)\,ds-t^{1/2}\left(\Gamma-\frac{\beta}{2}I\right)^{-1}T$$

for $t\in\mathbb{R}_0^+$ – one has

$$(2.78)\qquad \lim_{\substack{t\to\infty\\ t>0}}t^{-1/2+\sigma}\|[t]^{(1+\beta)/2}U_{[t+1]}-\hat{Y}(t)\|=0\quad a.s.$$

for any $\sigma\in[0,\min(\delta-\varepsilon,\eta,\gamma-\beta/2-\varepsilon-\eta))$.

Proof. (2.78) is an immediate consequence of Theorem 2.23 and (2.65).

Remark 2.25. (a) Some of the references quoted in Remark 2.18 also contain invariance principles of the form (2.24.C6). Further results of this type have been published by Komlós, Major and Tusnády [24], [25], Major [31] and Dehling [9].

(b) The main results of Ruppert's [46] paper, except for the case $\gamma=2\tau\neq 1/2-\gamma$ of Theorem 3.1 and Corollary 3.1 (which, however, is covered by our Proposition 2.1), are special cases of either Theorem 2.23 or Corollary 2.24.

The following corollary includes the (more important) case $c\alpha>1/2$ of Kersting's [23] Theorem 1 and, neglecting some technical problems connected with the explicit construction of the sequence (G_n) (below), its Hilbert space analogue established by Mark [32], Satz 3.3.

Corollary 2.26 (Kersting, Mark). *Let $\mathbb{H}$ be a real separable Hilbert space (with norm $\|\cdot\|$) endowed with its Borel σ-field $\mathscr{H}$. Let $M\colon\mathbb{H}\to\mathbb{H}$ be a measurable*

function, Fréchet differentiable at 0, *such that* $M(0)=0$,

(2.26.C1) $\Gamma = M'(0)$ *satisfies* (0.2.C2) *for some* $\gamma > 1/2$, *and – for some* $a \in (0,2)$ –

(2.26.C3.a) $$\|M(x) - \Gamma x\| = O(\|x\|^{1+a}) \quad \text{as } \|x\| \to 0.$$

Let $(G_n)_{n\in\mathbb{N}}$ *be a sequence in* $\mathscr{M}(\mathbb{H}\times\Omega, \mathbb{H})$, *and let* $(\mathscr{G}_n)_{n\in\mathbb{N}\cup\{0\}}$ *be a sequence of independent sub-σ-fields of* $\mathscr{A}$. *Suppose:*

(2.26.C0) $U_1 \in \mathscr{M}(\Omega, \mathbb{H})$ *is* $\mathscr{G}_0$*-measurable and*

$$U_{n+1} = U_n - n^{-1} M(U_n) - n^{-1} G_n(U_n, \cdot) \quad \text{for } n \in \mathbb{N}.$$

(2.26.C3.b) $U_n \to 0$ *a.s. as* $n \to \infty$.

(2.26.C4.a) *For all* $n \in \mathbb{N}$, *the mappings* G_n *are* $\mathscr{H} \otimes \mathscr{G}_n$*-measurable.*

(2.26.C4.b) *For each* $x \in \mathbb{H}$, $(G_n(x,\cdot))_{n\in\mathbb{N}}$ *is a sequence of identically distributed random variables.*

(2.26.C4.c) $$\sup_{x\in\mathbb{H}} E\|G_1(x,\cdot)\|^2 < \infty; \quad EG_1(x,\cdot) = 0 \text{ for all } x \in \mathbb{H}.$$

(2.26.C6) *There is a* $b \in (0,2)$ *such that* $E\|G_1(x,\cdot) - G_1(0,\cdot)\|^2 = O(\|x\|^b)$ *as* $\|x\| \to 0$.

Then – writing $Y_{n+1} = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^{\Gamma} k^{-1} G_k(0,\cdot)$ *for* $n \in \mathbb{N}$ *– one has*

(2.79) $$\lim_{N\to\infty} N^{1/2+\sigma} \|U_{N+1} - Y_{N+1}\| = 0 \quad \text{a.s.}$$

for any $\sigma \in [0, \frac{1}{4}\min(2a, b, 2\gamma - 1))$.

Proof. By (2.26.C3.a), M can be written in the form $M(x) = \tilde{\Gamma}(x)\,x$ $(x \in \mathbb{H})$, where $\tilde{\Gamma} \in \mathscr{M}(\mathbb{H}, \mathscr{L}(\mathbb{H}, \mathbb{H}))$ satisfies

(2.80) $$\|\tilde{\Gamma}(x) - \Gamma\| = O(\|x\|^a) \quad \text{as } \|x\| \to 0$$

(see, e.g., Mark [32], p. 36). Thus

(2.81) $$U_{n+1} = (I - n^{-1}\Gamma_n)\,U_n - n^{-1} V_n$$

with $\Gamma_n = \tilde{\Gamma}(U_n)$ and $V_n = G_n(U_n, \cdot)$. By virtue of Theorem 2.23 (take $T_k = T = 0$ and $Z_k = G_k(0,\cdot)$), the relations

(2.82) $$\|\Gamma_n - \Gamma\| = o(n^{-\delta}) \quad \text{a.s. for any } \delta \in (0, a/2),$$

(2.83) $$\left\|\sum_{k=1}^{n} V_k\right\| = o(n^{1/2+\varepsilon}) \quad \text{a.s. for any } \varepsilon \in \mathbb{R}^+$$

and

(2.84) $$\left\|\sum_{k=1}^{n} (V_k - G_k(0,\cdot))\right\| = O(n^{1/2-\eta}) \quad \text{a.s. for any } \eta \in (0, b/4)$$

are sufficient for (2.79) to hold. Using (2.26.C4.a), (2.26.C4.c) and the properties of the sequences (U_n) and $(\mathscr{G}_n)$, it follows from Satz 5.3.22 in [17] that (V_n) is a

martingale difference sequence with uniformly bounded second absolute moments. The martingale convergence theorem for $\mathbb{H}$-valued random variables (cf. Métivier [36], Theorem 11.7) implies that for any $\theta \in (1/2, 1)$, $\sum_{k=1}^{n} k^{-\theta} V_k$ converges a.s. as $n \to \infty$, which by the Banach space analogue of Kronecker's lemma entails (2.83). Applying Proposition 2.1 with

$$J = \{N\}, \quad v_{k,N} = V_k(\omega), \quad z_{k,N} = 0, \quad \lambda(m) = \left(\frac{m}{N}\right)^{1/2+\varepsilon},$$

$$u_{n,N}^{(0)} = U_{n+1}(\omega), \quad y_{n,N} = 0, \quad \tau = \rho = 1, \quad \delta = 0, \quad \theta = 1/2 + \varepsilon$$

(where $\omega \in \Omega$ is arbitrarily fixed) and using (2.83), we infer that

$$\|U_{N+1}\| = O(N^{-1/2+\varepsilon}) \quad \text{a.s. for any } \varepsilon \in \mathbb{R}^+. \tag{2.85}$$

Since $\|\Gamma_n - \Gamma\| = \|\tilde{\Gamma}(U_n) - \Gamma\| = O(\|U_n\|^a)$ a.s. (by (2.80) and (2.26.C3.b)), this yields (2.82). To prove (2.84), we first observe that – by (2.26.C6) and (2.26.C4.b) – there are constants r, $c_{12} \in \mathbb{R}^+$ such that

$$E\|G_n(x, \cdot) - G_n(0, \cdot)\|^2 \leqq c_{12} \|x\|^b \quad \text{for all } n \in \mathbb{N} \text{ and all } x \in \mathbb{H} \text{ with } \|x\| \leqq r. \tag{2.86}$$

Let $\varepsilon, \varepsilon^* \in (0, 1/2)$ be fixed. By virtue of (2.85) there is a constant $c(\varepsilon, \varepsilon^*) \in \mathbb{R}^+$ such that

$$P\{\sup_{n \in \mathbb{N}} n^{1/2-\varepsilon} \|U_n\| \leqq c(\varepsilon, \varepsilon^*)\} \geqq 1 - \varepsilon^*.$$

For $k \in \mathbb{N}$, let

$$\Delta_k = (V_k - G_k(0, \cdot))\, \chi\{\|U_k\| \leqq c(\varepsilon, \varepsilon^*)\, k^{-1/2+\varepsilon}\}.$$

The sequence $(\Delta_k)_{k \in \mathbb{N}}$ is a martingale difference sequence and – for $k \geqq (r^{-1} c(\varepsilon, \varepsilon^*))^{1/(1/2-\varepsilon)}$ –

$$\begin{aligned} E\|\Delta_k\|^2 &\leqq c_{12} E(\|U_k\|^b \chi\{\|U_k\| \leqq c(\varepsilon, \varepsilon^*)\, k^{-1/2+\varepsilon}\}) \\ &= O(k^{b(\varepsilon - 1/2)}). \end{aligned} \tag{2.87}$$

Together with Hölder's inequality and the Hilbert space versions of the martingale convergence theorem and Kronecker's lemma, this yields

$$\left\| \sum_{k=1}^{n} \Delta_k \right\| = o(n^{d/2}) \quad \text{a.s. for any } d > 1 + b(\varepsilon - 1/2). \tag{2.88}$$

Since $P\left(\bigcup_{k=1}^{\infty} \{\Delta_k \neq (V_k - G_k(0, \cdot))\}\right) \leqq \varepsilon^*$ and $\varepsilon^* \in (0, 1/2)$ was arbitrarily prescribed, (2.88) implies that

$$\left\| \sum_{k=1}^{n} (V_k - G_k(0, \cdot)) \right\| = o(n^{d/2}) \quad \text{a.s. for any } d > 1 + b(\varepsilon - 1/2). \tag{2.89}$$

Recalling that $\varepsilon \in (0, 1/2)$ was also arbitrary, we arrive at (2.84).

Remark 2.27. Suppose $(H_x)_{x\in\mathbb{R}}$ is a family of distribution functions satisfying the hypotheses of Kersting's [23] Theorem 1. Let H_x^{-1} be defined by $H_x^{-1}(t) = \sup\{s\colon H_x(s)\leqq t\}$ $(t\in(0,1))$, let $(\eta_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables uniformly distributed on $(0,1)$, and let $U_1\in\mathscr{M}(\Omega,\mathbb{R})$ be a random variable independent of $(\eta_n)_{n\in\mathbb{N}}$. Taking $G_n(x,\cdot)=H_x^{-1}(\eta_n)$ and noting that, by Blum's [5] Theorem 1, Kersting's Conditions 1–4 imply a.s. convergence of the corresponding Robbins-Monro process with coefficients of the form c/n (where $c\in\mathbb{R}^+$), it becomes evident that the case $c\alpha>1/2$ of Kersting's theorem is indeed a special case of Corollary 2.26.

3. Auxiliary Results

The first two lemmas below are special cases of Chow's inequality (cf. [17], Satz 6.6.1) and the Birnbaum-Marshall inequality ([4], Theorem 5.1).

Lemma 3.1. *Let $\theta\in(0,1/2)$ be fixed, and let $(S_k)_{k\in\mathbb{N}}$ be a nonnegative submartingale. Then*

$$\sup_{r\in\mathbb{R}^+}(r\,P\{\max_{k\leqq n} k^{-\theta}S_k\geqq r\})=\begin{cases}O(n^{1/2-\theta} & \text{if } ES_k=O(k^{1/2}),\\ o(n^{1/2-\theta} & \text{if } ES_k=o(k^{1/2}).\end{cases}$$

Lemma 3.2. *Let $\theta\in(0,1/2)$ be fixed, and let $(S_t)_{t\in\mathbb{R}_0^+}$ be a nonnegative separable submartingale such that $ES_t=ct^{1/2}$ for some $c\in\mathbb{R}^+$ and all $t\in\mathbb{R}_0^+$. Then, for all $r,T\in\mathbb{R}^+$,*

$$r\,P\{\sup_{0<t\leqq T} t^{-\theta}S_t\geqq r\}\leqq c(1-2\theta)^{-1}\,T^{1/2-\theta}.$$

In the sequel we shall use the notation introduced at the beginning of Sect. 2. For $(k,n)\in\mathbb{N}^2$ with $k<n$, we define

$$g(k,n)=\sum_{l=k+1}^{n} l^{-1}\,\|\Gamma_l^{(0)}-\Gamma\| \quad\text{and}\quad h(k,n)=\exp(g(k,n)). \tag{3.1}$$

Lemma 3.3. *There exist constants $c_1,c_2\in\mathbb{R}^+$ such that for all $(k,n)\in\mathbb{N}^2$ with $k<n$ the following estimates hold:*

$$\|A_{k,n}^{(0)}\|\leqq c_1\left(\frac{k}{n}\right)^{\gamma}h(k,n), \tag{3.2}$$

$$\|A_{k,n}^{(0)}-A_{k,n}^{(1)}\|\leqq c_2\,g(k,n)\left(\frac{k}{n}\right)^{\gamma}h(k,n). \tag{3.3}$$

Proof. Let $c_0=2^{-1}\,\|\Gamma\|^2\,e^{\|\Gamma\|}$. To verify (3.2), we note that (by (0.2.C2))

$$\begin{aligned}\|I-l^{-1}\Gamma_l^{(0)}\|&\leqq\|I-l^{-1}\Gamma\|+l^{-1}\,\|\Gamma_l^{(0)}-\Gamma\|\\ &\leqq\exp(-\gamma l^{-1}+c_0\,l^{-2}+l^{-1}\,\|\Gamma_l^{(0)}-\Gamma\|)\end{aligned}$$

for all $l\in\mathbb{N}$. Thus, setting $\tilde{c}_1=\exp\left(c_0\sum_{l=1}^{\infty}l^{-2}\right)=\exp(c_0\,\pi^2/6)$,

$$\|A_{k,n}^{(0)}\|\leqq\tilde{c}_1\exp\left(-\gamma\sum_{l=k+1}^{n}l^{-1}+g(k,n)\right), \tag{3.4}$$

which implies (3.2). We now proceed to the proof of (3.3). Starting from the recursion formulas

$$A_{k,k}^{(i)}=I, \qquad A_{k,l}^{(i)}=(I-l^{-1}\Gamma_l^{(i)})A_{k,l-1}^{(i)} \qquad (i\in\{0,1\})$$

valid for all $(k,l)\in\mathbb{N}^2$ with $k<l$, we find that

$$A_{k,l}^{(0)}-A_{k,l}^{(1)}=(I-l^{-1}\Gamma)(A_{k,l-1}^{(0)}-A_{k,l-1}^{(1)})-l^{-1}(\Gamma_l^{(0)}-\Gamma)A_{k,l-1}^{(0)}.$$

Hence

$$\begin{aligned}(3.5)\qquad \|A_{k,n}^{(0)}-A_{k,n}^{(1)}\| &\leqq \sum_{j=k+1}^{n}\|A_{j,n}^{(1)}\|\,j^{-1}\,\|\Gamma_j^{(0)}-\Gamma\|\,\|A_{k,j-1}^{(0)}\| \\ &\leqq \tilde{c}_1^2 e^{\gamma}\exp\left(-\gamma\sum_{l=k+1}^{n}l^{-1}\right)\left\{\sum_{j=k+1}^{n}j^{-1}\cdot\right. \\ &\quad \left.\cdot\,\|\Gamma_j^{(0)}-\Gamma\|\exp\left(\sum_{l=k+1}^{j-1}l^{-1}\,\|\Gamma_l^{(0)}-\Gamma\|\right)\right\} \qquad \text{(by (3.4))},\end{aligned}$$

and the inequalities $xe^y\leqq e^{y+x}-e^y$ and $e^x-1\leqq xe^x$ show that the factor in { } is $\leqq g(k,n)\,h(k,n)$.

Lemma 3.4. *For all $k\in\mathbb{N}$, one has*

$$(3.6)\qquad \|B_k^{(0)}\|\leqq k^{-1-\rho}(\rho+\|\Gamma_{k+1}^{(0)}\|)$$

and

$$(3.7)\qquad \|B_k^{(0)}-B_k^{(1)}\|\leqq k^{-1-\rho}\,\|\Gamma_{k+1}^{(0)}-\Gamma\|.$$

Proof. Obvious.

For $i\in\{0,1,2\}$, $n\in\mathbb{N}\setminus\{1\}$ and $\theta\in\mathbb{R}_0^+$, let

$$(3.8)\qquad D_0(n,\theta)=\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta}\|(k+1)^{-\rho}A_{k+1,n}^{(3)}-k^{-\rho}A_{k,n}^{(3)}\|,$$

$$(3.9)\qquad D_1^{(i)}(n,\theta)=\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta}\|A_{k+1,n}^{(i+1)}\|\,\|B_k^{(i)}-B_k^{(i+1)}\|,$$

$$(3.10)\qquad D_2^{(i)}(n,\theta)=\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta}\|A_{k+1,n}^{(i)}-A_{k+1,n}^{(i+1)}\|\,\|B_k^{(i)}\|.$$

Lemma 3.5.

$$D_0(n,\theta)\leqq\|\Gamma-\rho I\|\,n^{-\rho}\int_{1/n}^{1}t^{\theta+\gamma-\rho-1}\,dt.$$

Proof.

$$\begin{aligned}D_0(n,\theta)&=n^{-\rho}\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta}\left\|\left(\frac{k+1}{n}\right)^{\Gamma-\rho I}-\left(\frac{k}{n}\right)^{\Gamma-\rho I}\right\| \\ &\leqq n^{-\rho}\,\|\Gamma-\rho I\|\int_{1/n}^{1}t^{\theta+\gamma-\rho-1}\,dt\end{aligned}$$

(by (0.2.C2) and the (Banach space) mean value theorem).

Lemma 3.6. *Suppose that* (2.1.C3) *is satisfied for some* $\delta\in[0,1)$, *put* $\kappa=\theta+\gamma-\rho-\delta$, *and let the sequences* (a_n) *and* (b_n) *be defined by*

$$a_n=\begin{cases}1 & \text{if } \max(\kappa,\delta)>0\\ 1+g(1,n)\,h(1,n) & \text{if } \max(\kappa,\delta)\leqq 0\end{cases} \tag{3.11}$$

and

$$b_n=n^{-\rho-\delta}\,a_n\int_{1/n}^{1} t^{\kappa-1}\,dt. \tag{3.12}$$

Then

$$b_n^{-1}\sum_{j=1}^{2} D_j^{(0)}(n,\theta)=\begin{cases}o(1) & \text{if } \kappa>0,\\ O(1) & \text{if } \kappa\leqq 0.\end{cases} \tag{3.13}$$

Proof. We have

$$D_1^{(0)}(n,\theta)=O\left(\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta+\gamma}k^{-\rho-1-\delta}\right) \tag{3.14}$$

(by Lemmas 3.3 and 3.4)

$$=O\left(n^{-\rho-\delta}\int_{1/n}^{1} t^{\kappa-1}\,dt\right)$$

and

$$D_2^{(0)}(n,\theta)=O\left(\sum_{k=1}^{n-1}\left(\frac{k}{n}\right)^{\theta+\gamma}g(k,n)\,h(k,n)\,k^{-1-\rho}\right) \tag{3.15}$$

(by Lemmas 3.3 and 3.4)

$$=O\left(n^{-\rho}\int_{1/n}^{1} t^{\theta+\gamma-1-\rho}\,g([nt],n)\,h([nt],n)\,dt\right).$$

Recalling the definitions involved, the case $\kappa\leqq 0$ of the lemma can be settled by combining (3.14) and (3.15). Let us consider the case $\kappa>0$. Then, modifying the arguments used in (3.14) by means of the Toeplitz lemma, it is easily seen that "O" in (3.14) can be replaced by "o". Finally, it follows from (3.1), (2.1.C3) and Lebesgue's dominated convergence theorem that the right-hand side of (3.15) is $=o(b_n)$.

Remark 3.7. Letting the sequence $(\Gamma_n^{(0)})$ be defined by $\Gamma_n^{(0)}=n(I-\exp(-n^{-1}\Gamma))$, one has $\|\Gamma_n^{(0)}-\Gamma\|=O(n^{-1})$, so that (3.13) remains true with $D_j^{(0)}(n,\theta)$ replaced by $D_j^{(1)}(n,\theta)$, provided $\delta\in[0,1)$.

Lemma 3.8. *There exist constants* c_3, c_4, $c_5\in\mathbb{R}^+$ *such that for all* $(k,n)\in\mathbb{N}^2$ *with* $k<n$ *the following estimates hold:*

$$\|A_{k,n}^{(3)}\|\leqq\left(\frac{k}{n}\right)^{\gamma},\qquad \|A_{k,n}^{(2)}-A_{k,n}^{(3)}\|\leqq c_3\,k^{-1}\left(\frac{k}{n}\right)^{\gamma}, \tag{3.16}$$

$$\|B_k^{(2)}\|\leqq c_4\,k^{-\rho-1},\qquad \|B_k^{(2)}-B_k^{(3)}\|\leqq c_5\,k^{-\rho-2}. \tag{3.17}$$

Proof. Obvious.

Using Lemma 3.8 instead of Lemmas 3.3 and 3.4, the proof of the following lemma is similar to the proof of Lemma 3.6.

Lemma 3.9. *There is a constant $c_6 \in \mathbb{R}^+$ such that*

$$\sum_{j=1}^{2} D_j^{(2)}(n,\theta) \leqq c_6 n^{-\rho-1} \int_{1/n}^{1} t^{\theta+\gamma-\rho-2}\, dt. \tag{3.18}$$

4. Proof of Proposition 2.1

Let the mapping $\Psi_N\colon \mathscr{P}_N \times \mathscr{M}(\mathbb{G}_N, \mathscr{L}(\mathbb{B},\mathbb{B})) \times \mathscr{M}(I_N, \mathbb{B}) \to \mathbb{R}_0^+$ be defined by

$$\Psi_N(K,A,x) = \max_{n\in K} \left\| \sum_{j=1}^{n} A(j,n)\, x(j) \right\|.$$

Then, in consequence of the identity

$$\begin{aligned}\Psi_N(K,A,x) = \max_{n\in K} \Bigg\| & A(n,n) \sum_{j=1}^{n} x(j) \\ & + \sum_{l=1}^{n-1} (A(l,n) - A(l+1,n)) \sum_{j=1}^{l} x(j) \Bigg\| \\ & \text{(partial summation)},\end{aligned}$$

$$\Psi_N(K,A,x) \leqq C_N(K,A,\lambda)\, \Delta_N(K,x,\lambda), \tag{4.1}$$

which together with the obvious inequality

$$\max_{n\in J} n^{\tau} \|u_{n,N}^{(0)} - y_{n,N}\| \leqq \Psi_N(J,A',v^{(N)}) + \Psi_N(J,A'',v^{(N)} - z^{(N)}) \tag{4.2}$$

implies (a). Taking into account that – for all $(k,n) \in \mathbb{N}^2$ with $k<n$ –

$$\begin{aligned} & k^{-\rho}(A_{k,n}^{(0)} - A_{k,n}^{(3)}) - (k+1)^{-\rho}(A_{k+1,n}^{(0)} - A_{k+1,n}^{(3)}) \\ & = \sum_{i=0}^{2} \{A_{k+1,n}^{(i+1)}(B_k^{(i)} - B_k^{(i+1)}) + (A_{k+1,n}^{(i)} - A_{k+1,n}^{(i+1)})\, B_k^{(i)}\},\end{aligned}$$

the proof of (b) consists essentially in inserting the estimates stated in Lemmas 3.3–3.9 into the definition of $C_N(J,A',\lambda)$ and $C_N(J,A'',\lambda)$. In this way, (2.6) follows from (2.3), Lemmas 3.6 and 3.9 and Remark 3.7; (2.7) follows from (2.3) and Lemma 3.5.

5. Appendix: A Weak Invariance Principle for Martingale Difference Arrays in a Hilbert Space

The theorem to be proved in this section is a rather straightforward generalization of Walk's [52] Theorem 2 (see also Jakubowski [22], Theorem C). Having the applications to stochastic approximation in mind, it appeared to be useful to replace the existence of finite absolute second moments required by the above authors by the almost sure finiteness of second absolute conditional moments.

Let $\mathbb{H}$ be a real separable Hilbert space with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$ ($\|x\|=\langle x,x\rangle^{1/2}$). $\mathbb{H}$ will always be assumed to be endowed with its Borel σ-field. Let $(k_n)_{n\in\mathbb{N}}$ be a sequence of nondecreasing, right-continuous functions k_n: $[0,1]\to\mathbb{N}\cup\{0\}$ such that $k_n(0)=0$ and $k_n(1)\geqq 1$, let $(X_{n,i})_{i\in\{1,\dots,k_n(1)\},n\in\mathbb{N}}$ be an array of $\mathbb{H}$-valued random variables on a probability space $(\Omega,\mathscr{A},P)$ and let $(\mathscr{F}_{n,i})_{i\in\{0,\dots,k_n(1)\},n\in\mathbb{N}}$ be an array of sub-σ-fields of $\mathscr{A}$ such that – for each n and $1\leqq i\leqq k_n(1)$ – $X_{n,i}$ is $\mathscr{F}_{n,i}$-measurable, $\mathscr{F}_{n,i-1}\subset\mathscr{F}_{n,i}$ and $E(\|X_{n,i}\|^2|\mathscr{F}_{n,i-1})<\infty$ a.s.

Theorem 5.1. *Suppose*

(5.1) $E(X_{n,i}|\mathscr{F}_{n,i-1})=0$ *a.s. for all* $i\in\{1,\dots,k_n(1)\}$, $n\in\mathbb{N}$.

Let $S\in\mathscr{L}(\mathbb{H},\mathbb{H})$ *be an S-operator, i.e., a self-adjoint, positive definite operator with finite trace. If – for all* $x\in\mathbb{H}$, $\varepsilon\in\mathbb{R}^+$, $t\in[0,1]$ –

$$\sum_{i=1}^{k_n(1)} E(\|X_{n,i}\|^2|\mathscr{F}_{n,i-1})\xrightarrow{P}\operatorname{trace} S \quad \text{as } n\to\infty, \tag{5.2}$$

$$\sum_{i=1}^{k_n(1)} E(\langle X_{n,i},x\rangle^2\chi(\{|\langle X_{n,i},x\rangle|\geqq\varepsilon\})|\mathscr{F}_{n,i-1})\xrightarrow{P}0 \quad \text{as } n\to\infty \tag{5.3}$$

and

$$\sum_{i=1}^{k_n(t)} E(\langle X_{n,i},x\rangle^2|\mathscr{F}_{n,i-1})\xrightarrow{P}t\langle Sx,x\rangle \quad \text{as } n\to\infty, \tag{5.4}$$

then the sequence of random elements R_n *in* $C_{\mathbb{H}}[0,1]$ *(equipped with the supremum norm) defined by* $R_n(t)=\sum_{i=1}^{k_n(t)} X_{n,i}$ *if* $k_n(t)>k_n(t-)$ *and by linear interpolation otherwise converges in distribution to a Brownian motion* W *in* $\mathbb{H}$ *with* $W(0)=0$ *and covariance operator* S *of* $W(1)$.

Proof. Let $c>\operatorname{trace} S$ be a constant. For $j\in\{1,\dots,k_n(1)\}$, $n\in\mathbb{N}$, write

$$V_{n,j}^2=\sum_{i=1}^{j} E(\|X_{n,i}\|^2|\mathscr{F}_{n,i-1})$$

and

$$Y_{n,j}=X_{n,j}\chi(\{V_{n,j}^2\leqq c\}).$$

Then – adapting a related argument from the book of Gänssler and Stute [17], p. 364ff. –

(a) $$\lim_{n\to\infty} P\left(\bigcap_{i=1}^{k_n(1)}\{X_{n,i}=Y_{n,i}\}\right)=1,$$

(b) $$E(Y_{n,i}|\mathscr{F}_{n,i-1})=0 \quad \text{a.s. for all } i\in\{1,\dots,k_n(1)\},\ n\in\mathbb{N},$$

and

(c) the array $(Y_{n,i},\mathscr{F}_{n,i})$ satisfies (5.2)–(5.4) with $X_{n,i}$ replaced by $Y_{n,i}$ and "$\xrightarrow{P}$" replaced by convergence in first mean.

On account of (a) it suffices to prove the assertion of the theorem for the array $(Y_{n,i}, \mathscr{F}_{n,i})$. Moreover, it is enough to do this in the special case of a finite dimensional Hilbert space. This can be realized in the following way:

Let $\{e_j : j \in \mathbb{N}\}$ be a complete orthonormal system of $\mathbb{H}$, and let the operators Π_N: $\mathbb{H} \to \mathbb{H}$ be defined by $\Pi_N(x) = \sum_{j=1}^{N} \langle x, e_j \rangle e_j$ (for $N \in \mathbb{N}$). Then, for every $\varepsilon \in \mathbb{R}^+$ there exists an $N = N(\varepsilon)$ such that

$$E(\|W(1) - \Pi_N W(1)\|^2) = \sum_{j=N+1}^{\infty} \langle S e_j, e_j \rangle \leqq \varepsilon \tag{5.5}$$

and (using (c))

$$\sum_{i=1}^{k_n(1)} E(\|Y_{n,i} - \Pi_N Y_{n,i}\|^2) \leqq \varepsilon. \tag{5.6}$$

In consequence of Doob's inequality for nonnegative submartingales (cf. Métivier [36], Theorem 10.12), (5.5) and (5.6) lead to

$$\begin{aligned} E(\max_{t \in [0,1]} \|W(t) - \Pi_N W(t)\|^2) &\leqq 4 E(\|W(1) - \Pi_N W(1)\|^2) \\ &\leqq 4\varepsilon \end{aligned} \tag{5.7}$$

and

$$\begin{aligned} &E\left(\max_{t \in [0,1]} \left\| \sum_{i=1}^{k_n(t)} (Y_{n,i} - \Pi_N Y_{n,i}) \right\|^2\right) \\ &\leqq 4 E\left(\left\| \sum_{i=1}^{k_n(1)} (Y_{n,i} - \Pi_N Y_{n,i}) \right\|^2\right) \\ &= 4 \sum_{i=1}^{k_n(1)} E(\|Y_{n,i} - \Pi_N Y_{n,i}\|^2) \leqq 4\varepsilon. \end{aligned} \tag{5.8}$$

So it remains to ensure that, for each $N \in \mathbb{N}$, the array $(\Pi_N Y_{n,i}, \mathscr{F}_{n,i})$ obeys the weak invariance principle with limit process $\Pi_N W$, i.e., we have to show that – with $\tilde{R}_n^{(N)}$ defined by $\tilde{R}_n^{(N)}(t) = \sum_{i=1}^{k_n(t)} \Pi_N Y_{n,i}$ if $k_n(t) > k_n(t-)$ and by linear interpolation otherwise –

(i) the finite dimensional distributions of the sequence $(\tilde{R}_n^{(N)}(\cdot))_{n \in \mathbb{N}}$ converge to the corresponding finite dimensional distributions of $\Pi_N W(\cdot)$

and

(ii) the sequence $(\tilde{R}_n^{(N)}(\cdot))_{n \in \mathbb{N}}$ is tight.

Using (b), (c) and the Cramèr-Wold technique, (i) is a simple consequence of the central limit theorem for real-valued martingale difference arrays (see, e.g., Gänssler and Stute [17], Satz 9.2.3). Since N is fixed, tightness can be proved componentwise, so that – apart from minor modifications – it follows, e.g., from the proof of Lemma 10.1.9 in [17].

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